

1

Electrostatics

In this chapter, we will learn about

- the two kinds of electric charges
- methods of charging and discharging
- the electrostatic force between charges
- properties of electric fields
- the electric field strengths around a point charge and between two parallel charged plates

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1.1 Electric charges

You may get an electric shock when touching a door handle on a cool dry day in winter. This phenomenon is related to **electrostatics**, which is the study of **electric charges** at rest, i.e. static charges.



Fig 1.1a *You may get an electric shock easily in winter.*

Ts may discuss the following questions with Ss.

1 Is there anything flowing between the door handle and the person during the electric shock?

2 Will such electric shock happen in summer?

Suggested answers:

1 Electrons flow between the handle and the person.

2 It rarely occurs in summer as the humidity is usually high. Water vapour carries charges away from a body and prevents charges from accumulating.

DSE goal

Examine evidence for two kinds of charges.

Realize attraction and repulsion between charges.

DSE exam

12(1A)Q24, 15(1A)Q21

Video

→ This video demonstrates the Hands-on activity.

1 Two kinds of charges

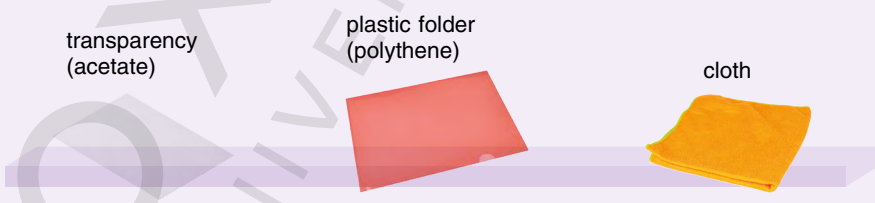
Let us learn more about charges in the following activity.

Attraction and repulsion

Hands-on activity

→ Worksheet is available on OUP web.

What you need

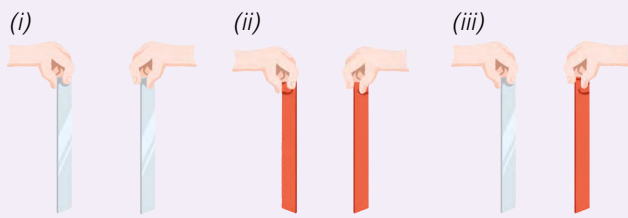


What to do

1 Cut long strips from the transparency and the plastic folder. Rub 2 strips with the cloth.



2 Hold the strips close together. Try the following cases: (i) 2 acetate strips, (ii) 2 polythene strips, (iii) 1 acetate strip and 1 polythene strip.



What you find

Do the strips attract or repel each other in each case?

The rubbed strips carry charges and exert **electrostatic force**, or **electric force**, on each other. Figure 1.1b on p.3 shows the results of the activity.

electrostatics 靜電學　electric charge 電荷　electrostatic force 靜電力　electric force 電力

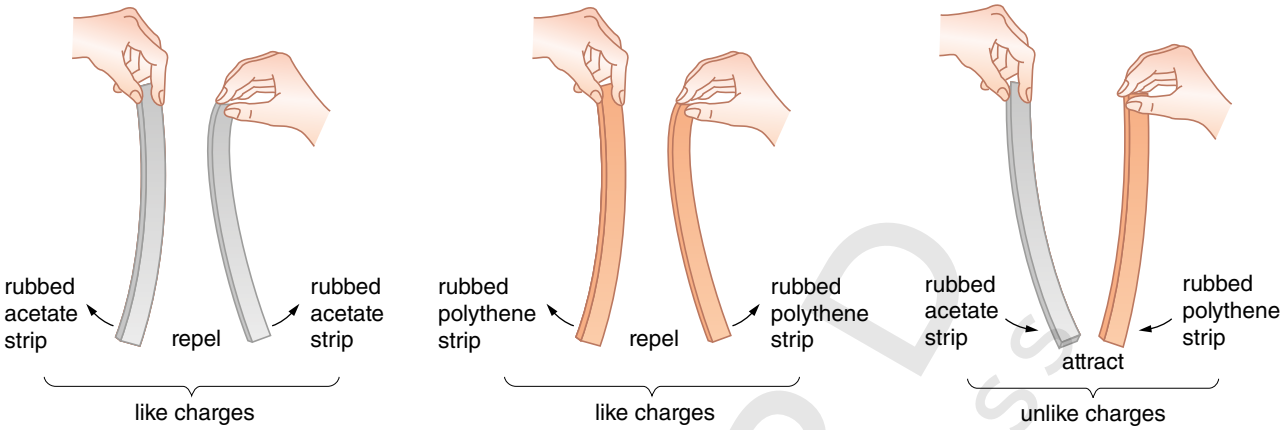


Fig 1.1b *Results of Hands-on activity.*

Know more

Two kinds of charges

French scientist *Charles du Fay* proposed that there are two kinds of charges, which were later called positive and negative by American scientist *Benjamin Franklin*.

The same kind of strips must carry the same kind of charge and they repel each other. However, strips of different kinds attract each other. This difference in interaction suggests that the acetate strip and the polythene strip carry different kinds of charges.

Scientists found that there are two kinds of charges, namely positive (+) and negative (–) charges. They concluded that:

Like charges repel; unlike charges attract.

When two charged objects are brought closer together, the electrostatic force between them will become greater.

2 Where do charges come from?

Cross-link

Positive charges in nuclei are due to the presence of protons. More details about the structure of an atom will be discussed in Book 5 Ch 2.

Recall that matter is made up of tiny particles called *atoms*. In the centre of each atom, there is a positively charged *nucleus*. Orbiting round the nucleus are the negatively charged **electrons** (Fig 1.1c). Normally, the positive and negative charges balance each other out in an atom. Therefore, the atom has no net charge and is said to be **neutral**.

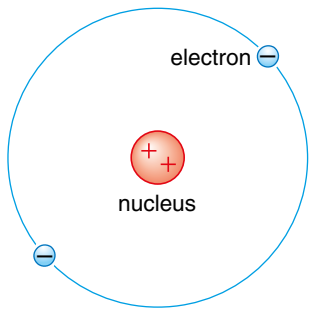


Fig 1.1c *A neutral atom.*

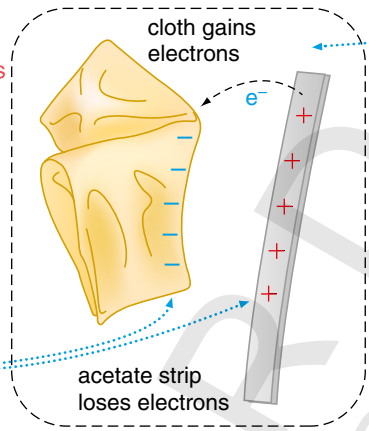
✎ Some Ss may wrongly think that an electron is a kind of charge. Ts should stress that electrons carry charge.

In many processes such as rubbing, a neutral object will lose electrons and become positively charged while another neutral object will gain extra electrons and become negatively charged (Fig 1.1d on p.4). Charges are only transferred between objects but not created in an isolated system.

The net charge of an isolated system is conserved.

Charles du Fay 篤費　Benjamin Franklin 富蘭克林　atom 原子　nucleus 原子核　electron 電子　neutral 中性

This is the *law of conservation of charge*. Note that only electrons but **not the nuclei** will be transferred between objects.



Remind Ss that protons are strongly bound inside the nucleus and can hardly be removed. Only electrons are transferred between objects.

charges are transferred but not created; net charge is conserved

The nuclei of some materials hold their electrons with greater attractive forces than other materials. That is why one material (larger attractive force) gains electrons while the other loses electrons after rubbing.

objects become oppositely charged and attract each other

Stress that the two objects carry the same amount of opposite charges.

Fig 1.1d *Electrons are transferred from one object to another in rubbing.*

The SI unit of charge is the **coulomb**, written as **C**. The charge of an electron is -1.60×10^{-19} C and this value is represented by e.

Know more

Coulomb

The coulomb is a very large unit.

When an acetate strip is rubbed with a cloth, about -10^{-9} C of charge (or 6×10^{9} electrons) are transferred to the cloth.

In a typical flash of lightning, about 10 C is transferred.



Smaller units such as µC and nC are usually used.

1 µC = 10^{-6} C

1 nC = 10^{-9} C

Quick check 1

→ E-version in OUP Exercise Platform is also available.

True or false:

1 A neutral object must not contain any charged particles. (T/F)

2 According to the law of conservation of charge, the amount of charge in an object is always conserved. (T/F)

3 An object becomes positively charged after being rubbed because positively charged particles are transferred to it from the cloth. (T/F)

Google Forms Microsoft Forms

→ This simulation shows, in stages, how a charged rod attracts (a) an aluminium foil and (b) a paper scrap by inducing charges on them. It is available on OUP web in PC version (.exe) only.

Simulation

3 Attraction of neutral objects

You must have seen this before: a rubbed plastic ruler attracts small neutral objects such as bits of metal foil or paper (Fig 1.1e). This suggests that:

A neutral object is attracted by a charged object.

Tip

Attraction exists between

1 objects with unlike charges,

2 a charged object and a neutral object.

Repulsion only exists between objects with like charges.

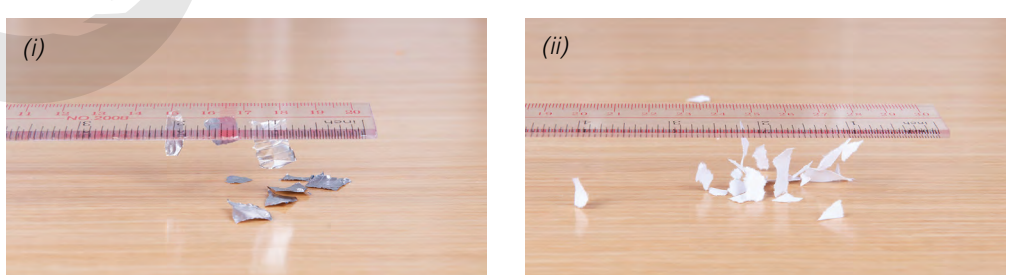


Fig 1.1e *A rubbed plastic ruler attracts small bits of (i) metal foil, (ii) paper.*

law of conservation of charge 電荷守恆定律 coulomb (C) 庫倫

Metal foil contains electrons that can move about freely. These electrons are called **free electrons**. An object with free electrons is an electrical **conductor**. On the contrary, paper is an electrical **insulator**, in which all electrons are tightly bounded to nuclei and cannot move about.

Learn more about conductors and insulators on p.53. ▶

Neutral conductors and insulators are attracted by charged objects through slightly different mechanisms (Fig 1.1f).

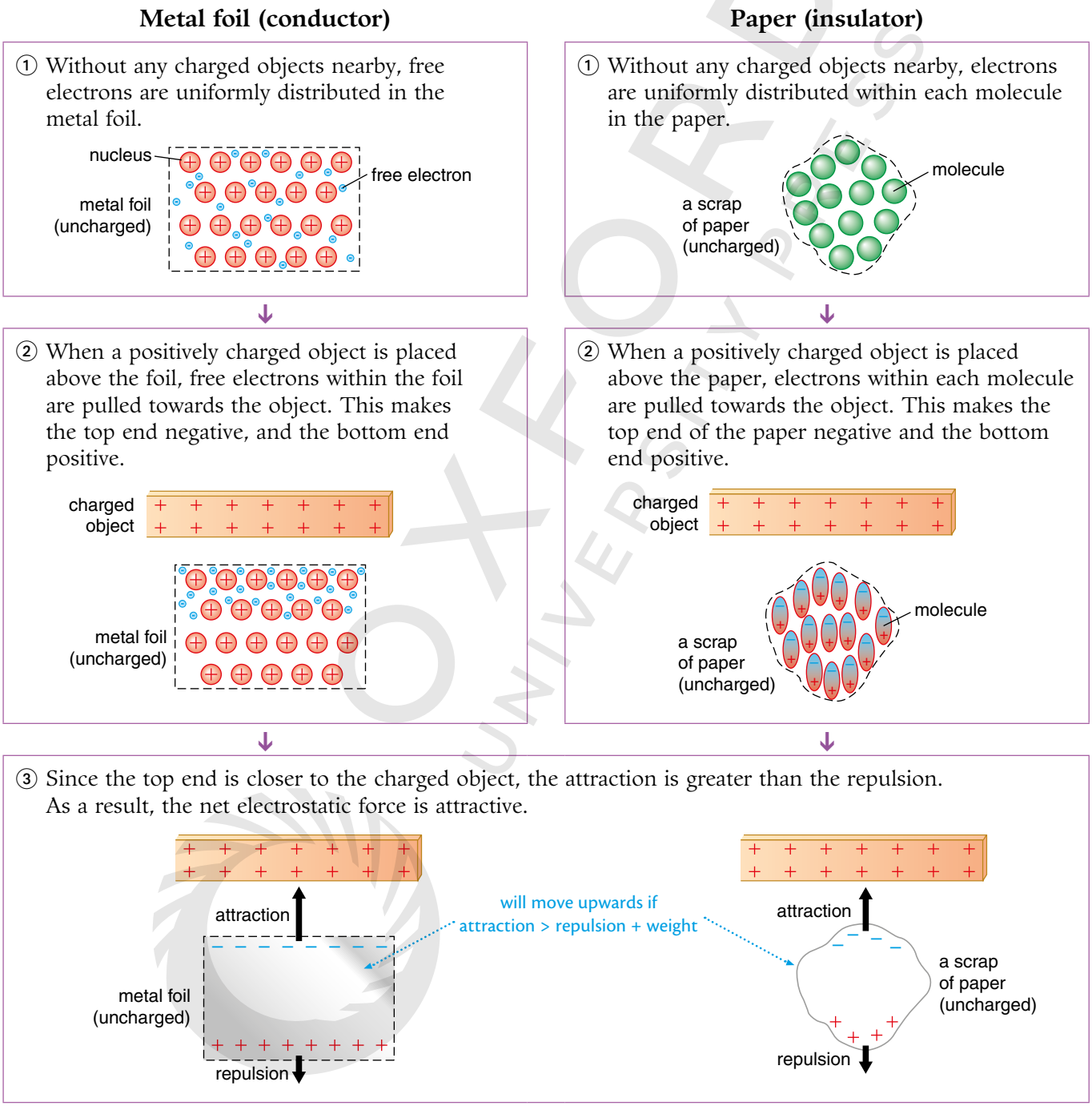


Fig 1.1f *How a charged object attracts a neutral conductor and insulator.*

When a charged object is placed nearby, charges inside a neutral object redistribute. This process is called **electrostatic induction** . The opposite charges that appear at the two ends are called **induced charges**.

free electron 自由電子　conductor 導體　insulator 絕緣體　electrostatic induction 靜電感應
induced charge 感生電荷

Video

→ This video demonstrates this phenomenon.

Writing practice

A plastic bag becomes negatively charged after it is rubbed by a cloth (Fig a). The charged bag is then put near a neutral metal can without touching (Fig b).

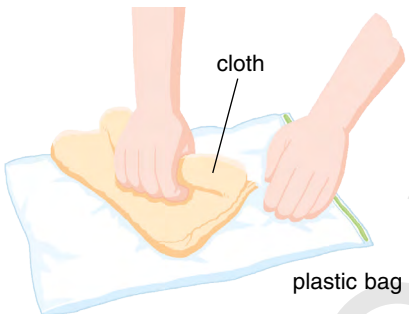


Fig a

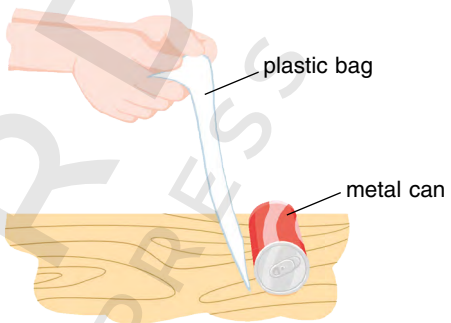


Fig b

Question 1

Explain why the plastic bag becomes negatively charged after it is rubbed by the cloth. (1 mark)

Answer

Some electrons are transferred from the cloth to the plastic bag. 1A

Question 2

Describe and explain the direction of the net electrostatic force acting on the metal can by the plastic bag. (2 marks)

Answer

On the can, positive charges are induced at the end near the bag and negative charges at the end far from the bag. 1A

The attraction on the near end is greater than the repulsion on the far end. Therefore, the net electrostatic force acts towards the bag. 1A

Tip

Some free electrons in the can are repelled by the bag to the far end.

Practise yourself

A plastic rod is rubbed by a cloth and becomes positively charged. It is then put near small bits of paper that are neutral.

1 Explain why the plastic rod becomes positively charged after it is rubbed by the cloth. (1 mark)

2 Describe and explain the direction of the net electrostatic force acting on the bits of paper by the plastic rod. (2 marks)

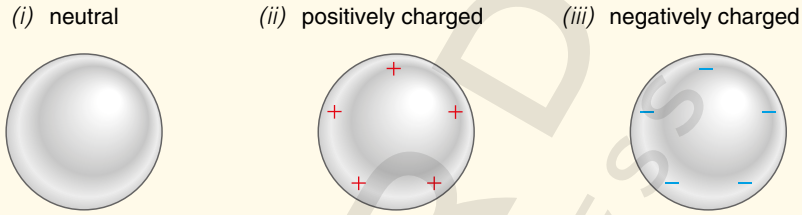
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→ Teaching notes and exercises are available on OUP web.

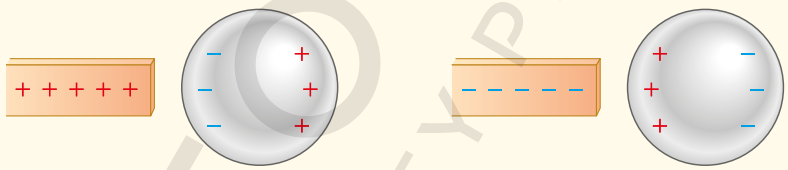
Skill
Drawing

Drawing charges on a metal sphere

1 No charged objects nearby: No need to draw any charges on a neutral sphere. Since like charges repel, draw uniformly distributed charges on the surface of a charged sphere.



2 A charged object nearby: In a neutral sphere, an **equal amount** of opposite charges are induced at the opposite ends.



Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) Charges can be induced in both conductors and insulators. (T/F)

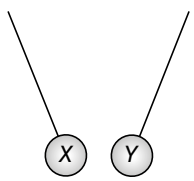
(b) When a negatively charged rod is put near a neutral conductor on a wooden table, a net positive charge is induced in the conductor. (T/F)

2 In the diagram, *X* and *Y* are balls suspended by light strings. Determine whether the following statements are true.

(a) *Y* may be neutral. (T/F)

(b) *X* and *Y* may carry like charges. (T/F)

(c) The balls must be conductors. (T/F)

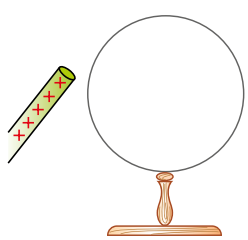


Learn by practice 1

1 A positively charged rod is placed near a neutral metal sphere without touching it.

(a) Sketch the induced charges on the sphere.

(b) Describe and explain the direction of the net electrostatic force acting on the sphere by the rod.



DSE goal

Interpret charging in terms of electron transfer.

DSE exam

14(1A)Q20, 21(1A)Q21

Flipped classroom

Watch a video and answer the questions.

4 Charging and discharging

Charging is the process by which a neutral object becomes electrically charged. **Discharging** is the reverse process by which an electrically charged object becomes neutral.

a Charging by friction

We already know that two neutral objects of different materials can be charged oppositely through rubbing. This is due to the transfer of electrons from one object to another (Fig 1.1d on p.4). This explains why your comb attracts your hair when you comb in winter (Fig 1.1g).

This charging method is called *charging by friction* and applies to **insulators** and **insulated conductors**.

An insulated conductor is a conductor separated from other conductors by insulator(s) so that charges cannot be transferred between it and other conductors.

Static charges are developed by friction between our shoes and the floor when we walk, and between the wheels and the road when a car travels.



Fig 1.1g Plastic comb attracts hair when combing.

b Charging by sharing

Figure 1.1h shows how a neutral metal sphere *B* is charged when it is touched by a negatively charged metal sphere *A*. This method is called *charging by sharing* and is only applicable to **conductors** as it involves the flow of free electrons.

If *A* is positively charged, some electrons will flow from *B* to *A* during touching and the positive charges are distributed between the spheres.

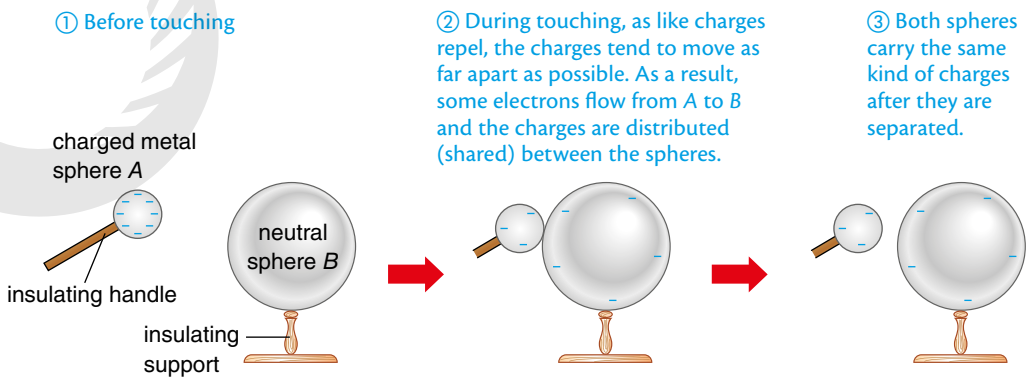


Fig 1.1h Charging a neutral conductor by sharing.

STSE

Air quality

Small particulates can be charged when they touch a charged object. This can be used to remove particulate pollutants in air.

▸ p.53

charging 起電　discharging 放電　charging by friction 摩擦起電　charging by sharing 授受起電

Note that the larger sphere *B* shares more charges. If the spheres have the same size, the charges will be equally shared between them. In either case, the **total charge** on the two spheres **conserves**.

Example 1 | Sharing charge

Identical conductors *X* and *Y* carry charges +5 μC and +1 μC respectively. *X* touches *Y* momentarily.

(a) Find the charge on each conductor after touching.

(b) State the particles that flow between *X* and *Y* and the direction of flow during touching.

Solution

Tip
Total charge is conserved. ▶

(a) Total amount of charge = 5 + 1 = 6 μC

Charge of each conductor = $\frac{6}{2}$ = 3 μC

(b) Electrons flow from *Y* to *X*.

C Discharging by earthing

The Earth is a conductor. It is always neutral. ▶

A special type of sharing of charges is called **earthing**, i.e. connecting a conductor to the Earth (e.g. via a metal wire) (Fig 1.1i). Since the Earth is so much larger than the conductor, any charges on the conductor are all distributed to the Earth and the conductor is discharged.

Note that the symbol ⏚ means earthing.

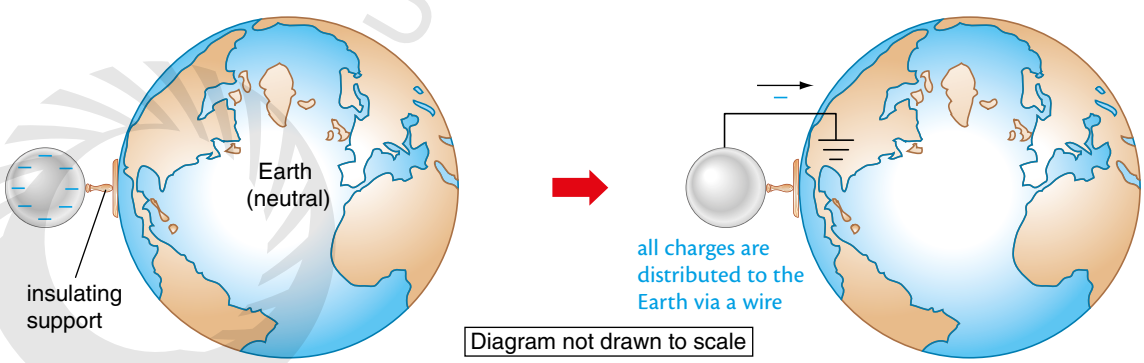


Fig 1.1i *In earthing, the net charge of a charged conductor is shared to the Earth (in the absence of other charged objects).*

The human body is a fairly good conductor. If you touch a charged conductor, you may connect it to the Earth so that it is discharged. This explains why insulating handle and support are needed in Figure 1.1h (p.8) so that the spheres will not be earthed.

earthing 接地

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 Neutral insulator *X* is rubbed by neutral insulator *Y*. After rubbing, *X* becomes positively charged. Determine whether the following statements are true.

(a) *X* and *Y* carry charges of the same magnitude. (T/F)

(b) *X* and *Y* attract each other. (T/F)

2 A positively charged conductor is earthed in the absence of other charged objects. Determine whether the following statements are true.

(a) Electrons flow from the conductor to the Earth. (T/F)

(b) The conductor becomes neutral. (T/F)

3 *True or false:* An insulator has no free electrons, so it cannot be charged. (T/F)

4 *X* and *Y* are two identical metal spheres. In the following cases, find the charges on each sphere before and after they touch each other.

	Before		After	
	X / μC	***Y*** / μC	***X*** / μC	***Y*** / μC
(a)	−6	0		
(b)		+10		+6
(c)	−6		+2	

Learn by practice 2

1 A cat plays in a box of foam peanuts. When it comes out of the box, foam peanuts are stuck on its body (Fig a). Explain this phenomenon briefly.



Fig a

2 In Figure b, draw the charge distribution in the spheres in steps ② and ③.

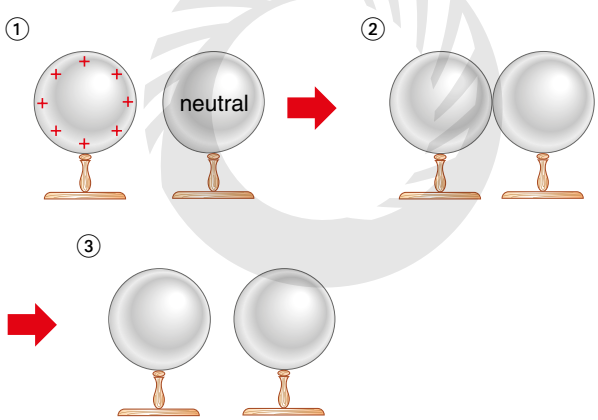


Fig b

3 *P* and *Q* are two identical metal spheres carrying charges of −8 μC and +4 μC respectively. Each of them is supported by a wooden stand.

(a) If the two spheres are brought into contact, what is the amount of charges flowing through? In which direction does the charge flow?

(b) Find the number of electrons that flow between the two spheres. The charge of an electron is -1.60×10^{-19} C.

(c) Why are *P* and *Q* held on a wooden stand?

4 A rubbed plastic ruler attracts bits of paper but a rubbed metal ruler does not (Fig c). Why?

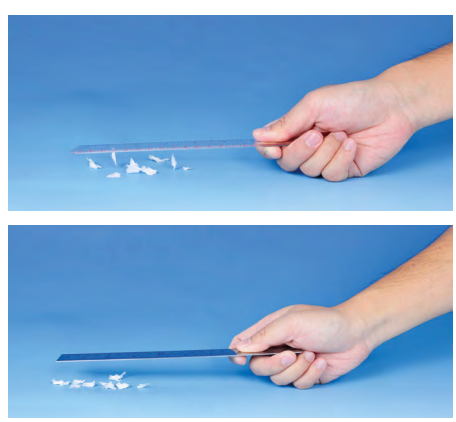


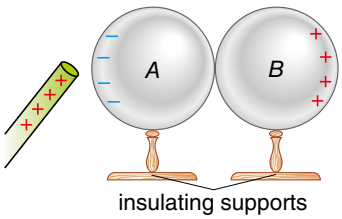
Fig c

d Charging by induction

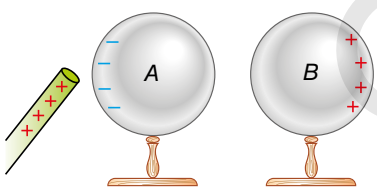
Do you still remember electrostatic induction? We can make use of it to charge an object. The method is called *charging by induction* (Fig 1.1j and 1.1k). Like charging by sharing, it involves the flow of free electrons, and so is only used for charging **conductors**.

Charging two metal spheres by induction

① Put neutral metal spheres *A* and *B* in touch with each other. Hold a charged rod near *A* without touching it. This induces opposite charges in *A* (close end) and *B* (far end).



② Separate the spheres with the rod held in position. The charges are held by the rod during the separation.



③ Remove the rod. The spheres carry opposite charges.

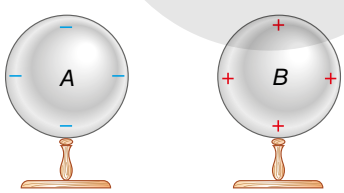
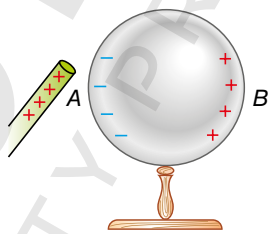


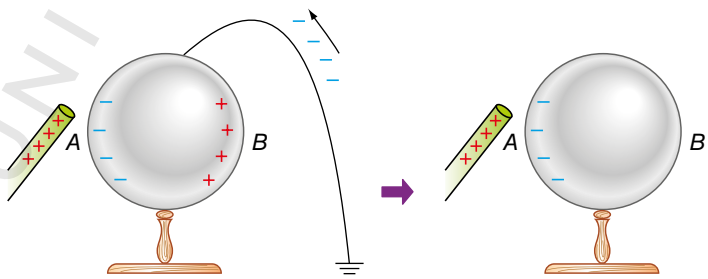
Fig 1.1j *Charging by induction.*

Charging a metal sphere by induction and earthing

① Hold a charged rod near a neutral metal sphere without touching it. This induces opposite charges at the close end *A* and the far end *B*.



② Earth the sphere momentarily with the rod held in position. Charges at the close end *A* are held by the rod. The flow of electrons during earthing neutralizes the charges at the far end *B*.



③ Remove the rod. The sphere carries charges.

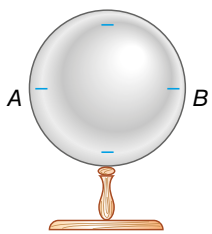


Fig 1.1k *Charging by induction and earthing.*

charging by induction 感應起電

Know more

Electric shock in winter

Static charges can cause electric shock and even form sparks.

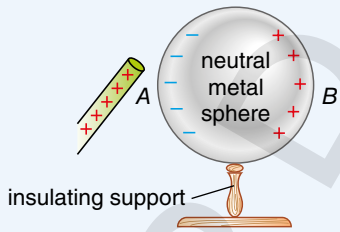


▸ p.54

Right and wrong

☆ HKDSE 2014 Paper 1A Q20

Which part of the sphere should be connected to the Earth so that the sphere becomes negatively charged?



✗ Region *B* must be connected because it is the position where the positive charges accumulate.

✓ We can connect any point on the sphere to the Earth.

No matter which part is connected to the Earth, the sphere and the Earth can be considered as two conductors in contact. The sphere is always the end near the rod and the Earth is the end far from the rod. Therefore, negative charges are induced on the sphere and positive charges are induced on the Earth.

Example 2 Flying aluminium foil

An activity involving different charging methods is shown below.

① Fix a styrofoam cup to the middle of a metal tray.



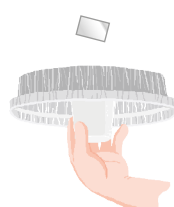
② Rub a styrofoam board with a cloth for over 30 s.



③ Place the tray on the board and momentarily touch the edge of the tray with your finger.



④ Hold the cup to turn the tray upside down. Release a small piece of aluminium foil onto the tray.



(a) The styrofoam board becomes charged in step ②. Which charging method is used?

(b) Explain why the metal tray becomes charged after it is touched momentarily in step ③.

(c) Explain why the foil 'flies' in the air after it is released and touches the tray in step ④.

Solution

(a) Charging by friction

(b) When the tray is placed on the board, opposite charges are induced at the bottom and at the edge of the tray. When it is touched momentarily, charges at the bottom are held by the board and charges at the edge are distributed to the Earth. As a result, the tray becomes charged.

(c) When the foil touches the tray, some charges on the tray are distributed to the foil. The foil then carries like charge as the tray and is repelled upwards by the tray.

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5 Gold leaf electroscope

A *gold leaf electroscope* is an instrument to test the presence of electric charges (Fig 1.1l). We can use it to demonstrate the charging and discharging methods described above.

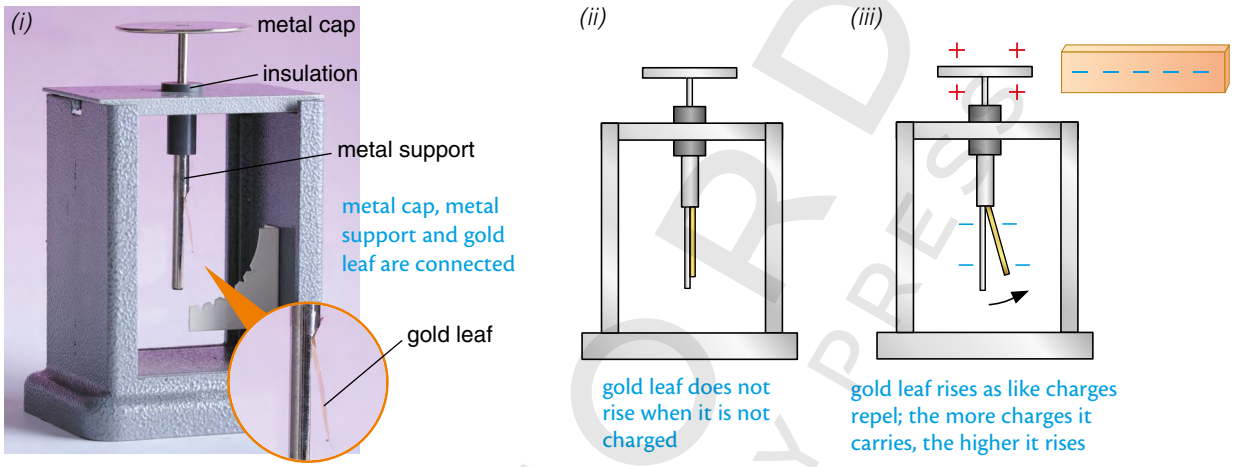


Fig 1.1l *A gold leaf electroscope.*

Video

→ This video demonstrates Expt 1a.

→ This is a Video Quiz about this experiment.

Experiment 1a Charging and discharging

Charging by sharing

1 Hold one end of a copper rod with a cloth. Rub the rod with a woolen cloth and then bring the rod in contact with the metal cap of the electroscope (Fig a). See what happens to the gold leaf.

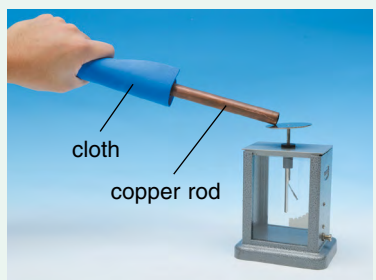


Fig a

Discharging by earthing

2 Right after step 1, touch the metal cap with your finger momentarily. See what happens to the gold leaf.

Charging by induction and earthing

3 Rub a plastic rod with a woolen cloth. Put the rod near the metal cap. Touch the metal cap with your finger momentarily while the rod is still being held in position (Fig b). Remove the rod. See what happens to the gold leaf.



Fig b

Discussion

1 Why must the copper rod in step 1 be held with a cloth?

2 Explain what happens to the gold leaf in each step.

gold leaf electroscope 金箔驗電器

Video

→ This video demonstrates experiments using Van de Graaff generator.

A school Van de Graaff generator can hold 1–2 μC charges. The voltage developed can be as high as 100 kV.

The charges built up on the dome can be positive or negative, depending on the material of the rubber belt.

Never earth the dome with your body. The large amount of charges may be harmful if they flow through your body.

→ This is a Video Quiz about an experiment using a Van de Graaff generator.

Example 3 | Van de Graaff generator

When a *Van de Graaff generator* (Fig a) is turned on, a large amount of charges accumulates on the dome.

(a) A pile of aluminium foil cake cups are put on the dome (Fig b). What happens to the cups after the generator is turned on? Explain.

(b) A neutral metal-coated foam ball suspended from an insulating rod is brought near the dome (Fig c). What happens to the ball after the generator is turned on? Explain.

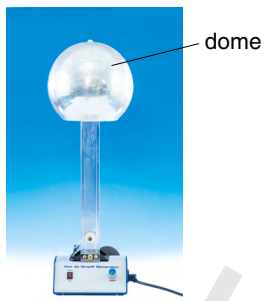


Fig a



Fig b



Fig c

Solution

(a) The cups fly away because they get the same kind of charges from the dome by sharing.

(b) The ball is first attracted to the dome because charges are induced on it. It is then repelled by the dome after touching the dome because it gets the same kind of charges from the dome by sharing.

Think about...

If the ball in (b) does not have the metal-coating, what will happen?

Quick check 4

→ E-version in OUP Exercise Platform is also available.

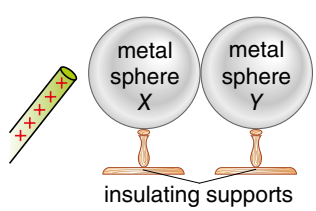
Google Forms | Microsoft Forms

1 A positively charged rod is brought close to two identical neutral metal spheres *X* and *Y* which are in contact with each other. Determine whether the following statements are true.

(a) *X* and *Y* carry the same kind of charges. (T/F)

(b) *X* and *Y* carry the same amount of charges. (T/F)

(c) If the rod is removed before *X* and *Y* are separated, *X* and *Y* will carry opposite charges. (T/F)

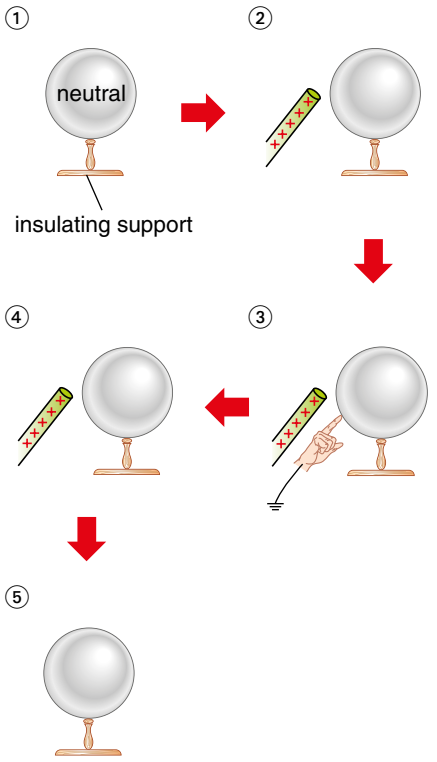


Van de Graaff generator 范德格拉夫起電機

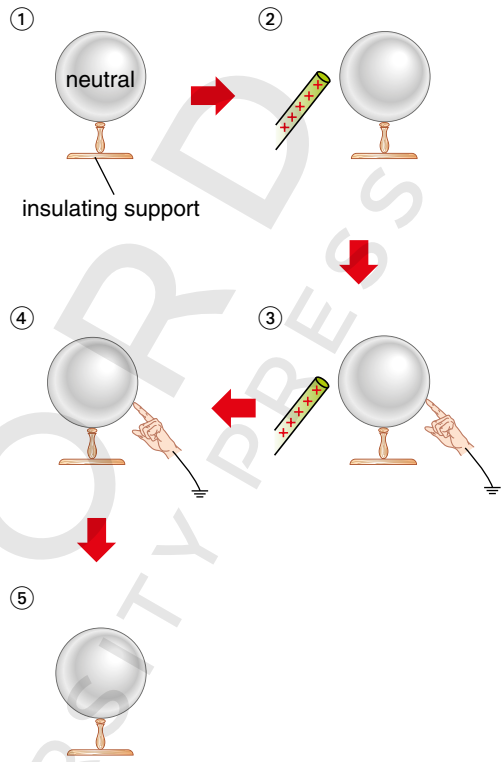
Learn by practice 3

Draw the charge distribution in the spheres (except those in step ①).

1



2



Exercise 1.1

(For Q1–2) A negatively charged plastic rod is brought near an insulated neutral metal rod (Fig a).

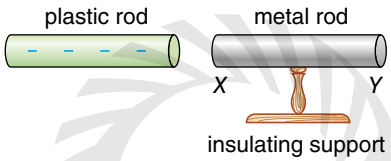


Fig a

1 Which of the following statements is correct?

A The metal rod becomes charged.

B Some nuclei in the metal rod are attracted to side *X*.

C All electrons in the metal rod are repelled to side *Y*.

D The plastic rod is attracted by the metal rod.

★**2** While the rods remain in position, side *X* is earthed momentarily. Which of the following will happen?

(1) Electrons move from the metal rod to the Earth.

(2) The metal rod becomes positively charged.

(3) Side *Y* remains negatively charged.

A (2) only B (3) only

C (1) and (2) only D (1) and (3) only

★**3** Conductor *X* is connected to the Earth and is brought near a positively charged metal sphere *Y* without touching. Which of the following statements is/are correct?

(1) *X* becomes negatively charged.

(2) Free electrons move to *Y*.

(3) *X* and *Y* attract each other.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

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★**4** *X*, *Y* and *Z* are charged plastic balls suspended by light strings (Fig b).

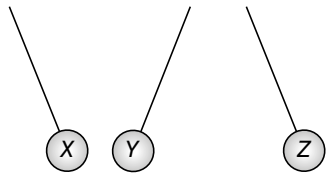


Fig b

Which of the following combinations is/are possible?

	X	*Y*	*Z*
A	+	+	−
B	+	−	−
C	−	−	+
D	−	+	−

★**5** *P*, *Q* and *R* are three metal spheres on insulating supports (Fig c). They are of different sizes. *P* is positively charged while *Q* and *R* are neutral. *P* is brought close to *Q* without touching. Then *R* touches the right side of *Q* momentarily.

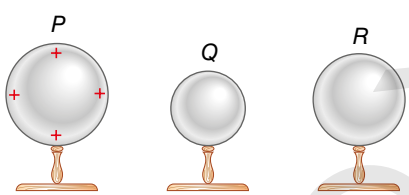


Fig c

(a) What happens when *R* is touching *Q*?

(b) Are *Q* and *R* positively or negatively charged after touching?

(c) After touching, *Q* carries a charge of magnitude 2 μC. Is the magnitude of the charge in *R* larger than, equal to or smaller than 2 μC?

★**6** *X*, *Y* and *Z* are three identical metal spheres on insulating supports and carry charges –2 μC, *Q* and +5 μC respectively. *Y* is brought to touch *X* and *Z* in turn. The final charge on *Z* is +2 μC.

(a) What is the final charge on *Y*?

(b) What is the charge on *Y* after touching *X* but before touching *Z*?

(c) Find *Q*.

(d) What will happen if the spheres are made of plastic instead of metal? Explain.

★**7** (a) Why does a metal rod not become charged when it is held by a bare hand and rubbed with a piece of cloth?

(b) Suggest a method to charge a metal rod.

★**8** Two metal-coated polystyrene spheres are hung from the ceiling with nylon threads. Both spheres are positively charged. Peter, wearing insulating gloves, holds the spheres at the positions as shown (Fig d).

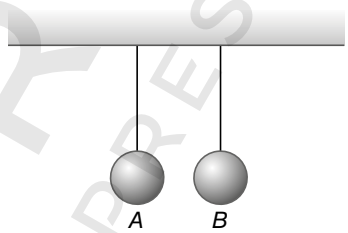


Fig d

The spheres are then released.

(a) Describe the motion of the spheres shortly after release.

(b) How does the result in (a) change if Mary accidently touches *B* with her bare hands before the spheres are released? Explain.

★**9** A girl stands on a plastic stool and puts her hands on a metal dome. After a large amount of charges are transferred to the dome, the girl's hair stands on end (Fig e).



Fig e

(a) Why does the girl's hair stand on end?

(b) Why should the girl stand on a plastic stool instead of on the ground?

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1.2 Electrostatic force

Figure 1.2a shows electrostatic forces in action: when a charged balloon is put near a stream of water, charges are induced in the water. Electrostatic forces exist between the charges in the two objects, pulling the balloon towards the water and bending the stream of water towards the balloon.



Fig 1.2a *Electrostatic forces exist between charges.*

In this part, we shall learn about the electrostatic force in a quantitative way.

DSE goal
Stating Coulomb's law.

1 Coulomb's law

In 1785, *Charles Coulomb* discovered that the magnitude of the electrostatic force F between two charges is directly proportional to the product of charges (Q_1 and Q_2), and inversely proportional to the square of the distance r between them (Fig 1.2b).

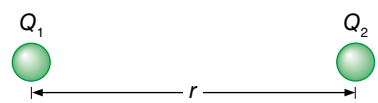


Fig 1.2b *Two point charges separated by distance r.*

This is known as **Coulomb's law**. Mathematically,

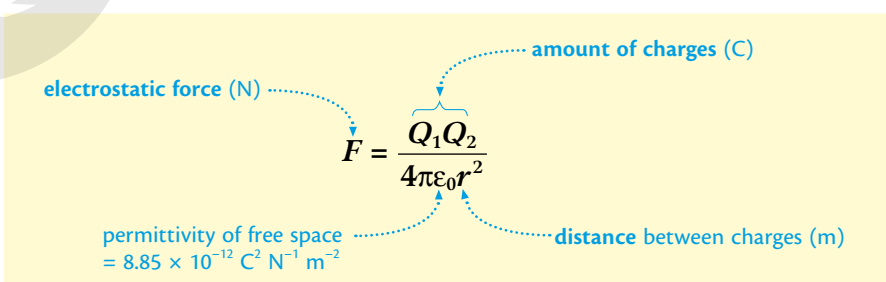
$$F = k \times \frac{Q_1 Q_2}{r^2}$$

where k is the proportionality constant called *Coulomb's constant* and is approximately equal to 8.99×10^9 N m^2 C^{-2}. k can be written as $\frac{1}{4\pi\varepsilon_0}$, where ε_0 is a constant called the *permittivity of free space* and is approximately 8.85×10^{-12} C^2 N^{-1} m^{-2}. Therefore,

ε_0 is pronounced 'epsilon nought' or 'epsilon zero'.

Free space is a theoretically 'perfect' vacuum. For simplicity, all charges are assumed to be in free space in this book.

You can work out the unit of ε_0 from this equation.



Charles Coulomb 庫倫　Coulomb's law 庫倫定律　Coulomb's constant 庫倫常數
permittivity of free space 真空電容率

DSE goal

Solving problems involving forces between point charges.

DSE exam

13(1A)Q24, 13(1B)Q11(a), 14(1A)Q21, 17(1A)Q22, 19(1A)Q23, 22(1A)Q23, 23(1A)Q23

2 Electrostatic force between charges

When finding the electrostatic force between two charges (Fig 1.2c) , note that:

1 Electrostatic force acts along the straight line joining the two charges. Like other forces, electrostatic force is a vector quantity.

2 Electrostatic force is repulsive if the charges have the same sign, and attractive if the charges have opposite signs.

3 Electrostatic force exists in pairs and obeys Newton's third law. They are **equal in magnitude** ($F_{12} = F_{21}$), but **opposite in directions**. They form an action-and-reaction pair.

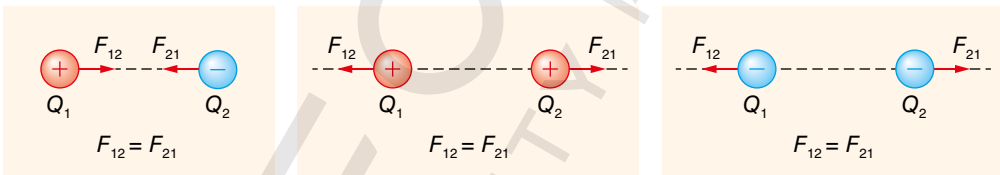


Fig 1.2c *Attractive force and repulsive force between charges.*

4 For two given charges (Q_1 and Q_2 remain unchanged), $F \propto \frac{1}{r^2}$. This is an example of inverse-square law (Fig 1.2d). Doubling the distance between the charges changes the force to $\frac{1}{4}$ of its original value.

Newton's law of universal gravitation $F = \frac{Gm_1m_2}{r^2}$ is another example of inverse-square law.

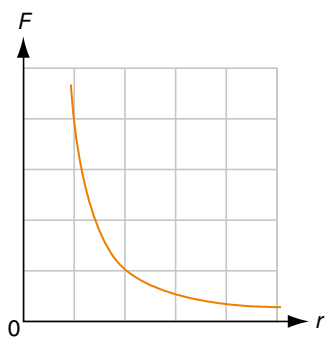


Fig 1.2d *The graph of F against r, showing their inverse-square relationship.*

Know more

Charles Coulomb (1736–1806)

Charles Coulomb made significant contributions to the study of mechanics, electricity and magnetism. He is most famous for his investigation into the force between electric charges. The unit of charge was named in his honour in 1880.



→ Teaching notes and exercises are available on OUP web.

Skill
Problem-solving

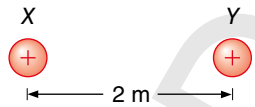
Finding electrostatic force

1 The direction of F can be determined from the types of charges involved: like charges repel; unlike charges attract.

2 The magnitude of F can be determined by applying Coulomb's law. Ignore the signs of the charges in the calculation.

Example 4 Electrostatic forces between point charges

Point charges X ($+2 \times 10^{-8}$ C) and Y ($+10^{-8}$ C) are 2 m apart.



Take $\varepsilon_0 = 8.85 \times 10^{-12}$ C^2 N^{-1} m^{-2}.

(a) Find the electrostatic force F_{XY} acting on X by Y.

(b) Find the electrostatic force F_{YX} acting on Y by X.

(c) Now, Y is replaced with Z (-10^{-8} C). Find the electrostatic force F_{XZ} acting on X by Z.

Solution

(a) $F_{XY} = \dfrac{Q_1Q_2}{4\pi\varepsilon_0 r^2} = \dfrac{(2 \times 10^{-8})(10^{-8})}{4\pi(8.85 \times 10^{-12})(2)^2} = 4.50 \times 10^{-7}$ N (to the left)

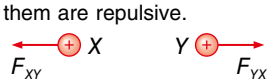
(b) F_{YX} is 4.50×10^{-7} N (to the right).

(c) F_{XZ} is 4.50×10^{-7} N (to the right).

Ts may ask Ss to use the Coulomb's constant k and check if they can get the same results.

Tip

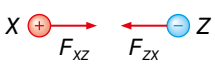
(a) As X and Y carry like charges, the forces between them are repulsive.



(b) F_{XY} and F_{YX} are an action-and-reaction pair.

(c) As Y and Z carry the same amount of charges, F_{XZ} and F_{XY} have the same magnitude.

As X and Z carry unlike charges, the forces between them are attractive.



Example 5 How F is related to Q and r

Point charges X and Y are r apart. The magnitude of the electrostatic force acting on X by Y is F_0. Find the magnitude of the electrostatic force F if

(a) the charge carried by X is doubled,

(b) the distance between X and Y is doubled.

Solution

Originally, $F_0 = \dfrac{Q_1Q_2}{4\pi\varepsilon_0 r^2}$

(a) As Q_1 is doubled,

$$F = \frac{(2Q_1)Q_2}{4\pi\varepsilon_0 r^2} = 2 \times \frac{Q_1Q_2}{4\pi\varepsilon_0 r^2} = 2F_0$$

(b) As r is doubled,

$$F = \frac{Q_1Q_2}{4\pi\varepsilon_0 (2r)^2} = \frac{1}{4} \times \frac{Q_1Q_2}{4\pi\varepsilon_0 r^2} = \frac{F_0}{4}$$

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 Complete the following table about the magnitude of the electrostatic force between two charges. Take $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$.

	Q_1	Q_2	r	F
(a)	2×10^{-6} C	3×10^{-6} C	0.5 m	N
(b)	7×10^{-7} C	4×10^{-6} C	m	6.29×10^{-3} N
(c)	nC	6×10^{-8} C	0.03 m	3×10^{-3} N
(d)	10 nC	nC	10 mm	0.0899 N

2 Are the following statements about two small charged objects true or false?

(a) If the distance between them is doubled, the electrostatic force between them will be halved. (T/F)

(b) If the amount of charge in one object is doubled, the electrostatic force between them will be doubled. (T/F)

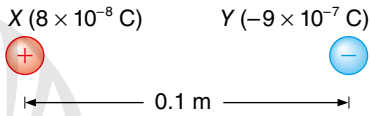
(c) The forces that they exert on each other have the same magnitude but in opposite directions. (T/F)

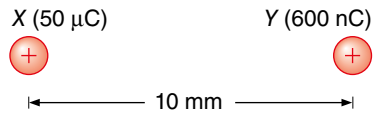
Learn by Practice 4

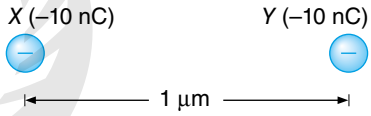
Take $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$.

1 Inside a hydrogen atom, the electron orbits the nucleus in a circular path. The charge of the electron is -1.6×10^{-19} C and the charge of the nucleus is $+1.6 \times 10^{-19}$ C. The magnitude of the electrostatic force between the electron and the nucleus is 8.23×10^{-8} N. Find the radius of the circular path.

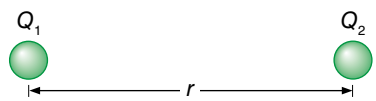
2 X and Y are two point charges. Find the magnitude and direction of the electrostatic forces they exert on each other in the following cases.

(a) 

(b) 

(c) 

3 Two point charges Q_1 and Q_2 are r apart. The force acting on Q_2 by Q_1 is F.



(a) If $Q_1 = 4 \times 10^{-10}$ C, $r = 0.01$ m and $F = 1.8 \times 10^{-4}$ N (to the right), find Q_2.

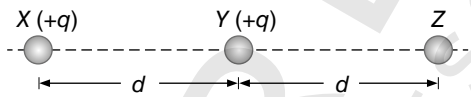
(b) If $Q_2 = 2$ μC, $r = 2$ mm and $F = 4500$ N (to the left), find Q_1.

Cross-link
You learned the vector sum in Book 2 Ch 4.

If more charges are involved, the resultant electrostatic force on any charge is the **vector sum** of individual forces on it.

Example 6 Finding the charge of a particle

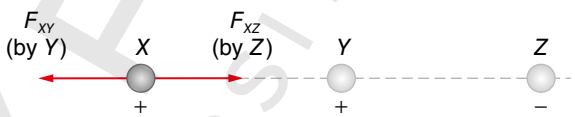
Charged particles X, Y and Z are placed on a straight line such that Y is at equal distance d from X and Z. X and Y each have a charge of $+q$.



If the resultant force acting on X due to Y and Z is zero, find the charge carried by Z.

Solution

The resultant force acting on X being zero implies that F_{XY} and F_{XZ} are in opposite directions. Therefore, Z is negatively charged.



Let the charge carried by Z be $-Q$.

$$F_{XZ} = F_{XY}$$

$$\frac{1}{4\pi\varepsilon_0}\frac{(q)(Q)}{(2d)^2} = \frac{1}{4\pi\varepsilon_0}\frac{(q)(q)}{(d)^2}$$

$$Q = 4q$$

$\therefore$ The charge carried by Z is $-4q$.

Think about...

How would the answer change if X has a charge of $+2q$?

→ Teaching notes and exercises are available on OUP web.

Skill *Problem-solving*

Finding resultant force

1. Draw a free-body diagram to show all the forces acting on the object.
2. Apply Coulomb's law to find the magnitude of each electrostatic force.
3. Forces can be resolved along two perpendicular directions. Find the resultant force in each direction. The magnitude of the resultant force F can be found by using Pythagoras' theorem. The direction of F can be found by using trigonometric function.

Example 7 Finding resultant electrostatic force

Point charges X ($+10^{-8}$ C), Y ($+10^{-8}$ C) and Z (-4×10^{-8} C) are placed at the three vertices of a right-angled triangle (Fig a). Find the direction and magnitude of the resultant electrostatic force acting on X by Y and Z.

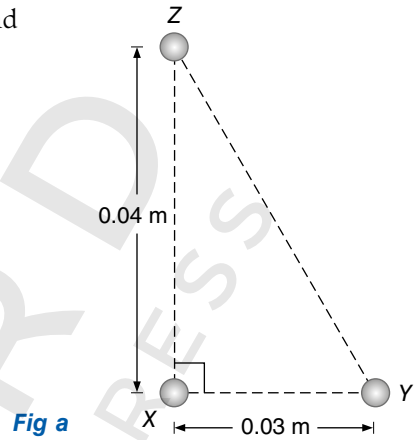


Fig a

Solution

The free-body diagram of X is as shown in Figure b.

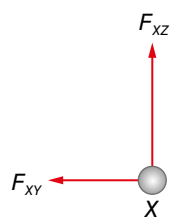


Fig b

Apply $F = \dfrac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$.

$$F_{XY} = \frac{(10^{-8})(10^{-8})}{4\pi(8.85 \times 10^{-12})(0.03)^2}$$

$$= 9.991 \times 10^{-4} \text{ N}$$

$$F_{XZ} = \frac{(10^{-8})(4 \times 10^{-8})}{4\pi(8.85 \times 10^{-12})(0.04)^2}$$

$$= 2.248 \times 10^{-3} \text{ N}$$

Magnitude of the resultant force F (Fig c)

$$= \sqrt{F_{XY}^2 + F_{XZ}^2}$$

$$= \sqrt{(9.991 \times 10^{-4})^2 + (2.248 \times 10^{-3})^2}$$

$$= 2.46 \times 10^{-3} \text{ N}$$

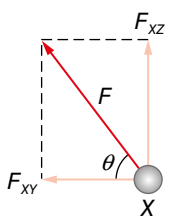


Fig c

For the direction of F,

$$\tan\theta = \frac{F_{XZ}}{F_{XY}}$$

$$= \frac{2.248 \times 10^{-3}}{9.991 \times 10^{-4}}$$

$$\theta = 66.0°$$

$\therefore$ The resultant force acting on X is 2.46×10^{-3} N towards the left at 66.0° above the horizontal.

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Quick check 6

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(For Q1–2.) *PQRS* is a straight line. Point charges *X* and *Y* are placed at points *Q* and *R* respectively (Fig a). On *PS*, a positive charge *q* experiences zero electrostatic force.



Fig a

1 In which region(s) may *q* lie if *X* and *Y* are both positive?

☐ *PQ* ☐ *QR* ☐ *RS*

2 In which region(s) may *q* lie if *X* is positive but *Y* is negative?

☐ *PQ* ☐ *QR* ☐ *RS*

3 *ACEG* is a square with centre *O*. *B*, *D*, *F* and *H* are the mid-points of its sides. Point charges *X* (+*q*), *Y* (–*q*) and *Z* (+*q*) are placed at points *B*, *H* and *O* respectively (Fig b). What is the direction of the resultant force acting on *Z* by *X* and *Y*?

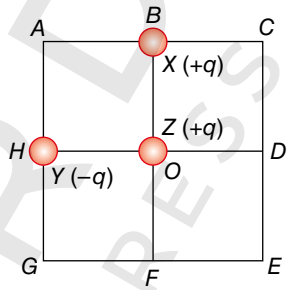


Fig b

A Along *OB* B Along *OH*

C Along *OA* D Along *OG*

Learn by practice 5

1 Two point charges *X* (–*Q*) and *Y* (+9*Q*) are 4 cm apart (Fig a). A charge *q* is placed at *A* and experiences an electrostatic force *F* from *X*.

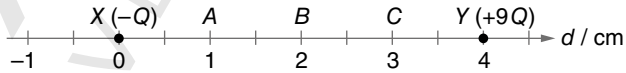


Fig a

Complete the table below by filling in the electrostatic force acting on *q* when it is placed at different positions. Give your answers in terms of *F*.

Position	Force from *X*	Force from *Y*	Resultant force
A	*F*		
B			
C			

2 Point charges *A*, *B* and *C* are placed at positions as shown (Fig b).

Find the magnitude and direction of the resultant force acting on *B* in the following cases.

(a) The force acting on *B* by *A* is 0.05 N (attractive) and that acting on *B* by *C* is 0.05 N (attractive).

(b) The force acting on *B* by *A* is 3×10^{-3} N (attractive) and that acting on *B* by *C* is 4×10^{-3} N (repulsive).

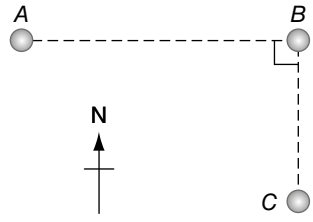


Fig b

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Exercise 1.2

1 Two point charges $+Q$ and $+2Q$ are 3 cm apart. A charge q is placed between them and at r cm from $+Q$ (Fig a).

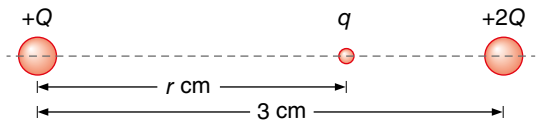


Fig a

If the charge q experiences no net force, which of the following equations about r is correct?

A $\frac{1}{r^2} = \frac{2}{(3-r)^2}$ B $\frac{1}{r^2} = \frac{2}{3^2}$

C $\frac{1}{3^2} = \frac{2}{(3-r)^2}$ D $\frac{1}{3^2} = \frac{2}{(3-r)^2}$

★**2** X, Y and Z are three point charges with the same amount of charge. Along which direction is the net electrostatic force acting on X (Fig b)?

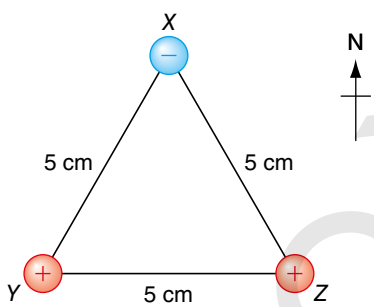


Fig b

A East
B South
C West
D North

★**3** Two point charges $-4Q$ and $-Q$ are 6 cm apart (Fig c). Where should a negative charge be placed so that the resultant electrostatic force acting on it is zero?

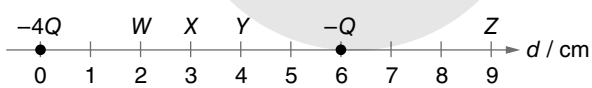


Fig c

A W B X

C Y D Z

★**4** Two identical balls X and Y are suspended by nylon strings. They carry the same amount of charge. Both threads make the same angle with the vertical (Fig d).

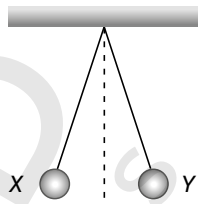
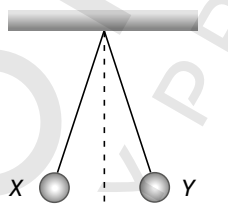


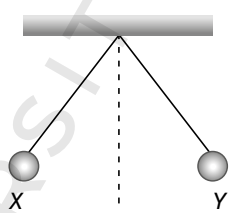
Fig d

If the quantity of charge in X increases, how will the angles between the threads and the vertical change?

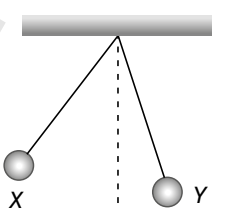
A



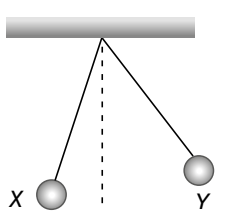
B



C



D



5 A negative point charge Y exerts an electrostatic force F of magnitude 5.5×10^{-8} N on another negative point charge X (Fig e). It is known that X and Y carry the same charges and the distance between X and Y is 0.03 m. Draw Y at a suitable position in Figure e and find its charge.

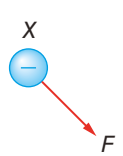


Fig e

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6 Two point charges X ($+5 \times 10^{-8}$ C) and Y (-2×10^{-8} C) are 10 cm apart (Fig f). A point charge Z (-3×10^{-8} C) is put at the mid-point between X and Y.

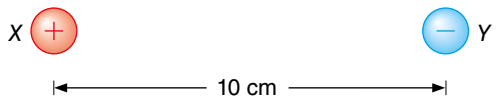


Fig f

(a) Find the magnitude of the electrostatic force acting on Y by Z.

(b) Find the net electrostatic force acting on Y.

★**7** Two point charges Q_1 (5×10^{-8} C) and Q_2 (-2×10^{-8} C) are placed at distance r apart (Fig g). The magnitude of the electrostatic force F acting on Q_1 by Q_2 is 5.5×10^{-8} N.

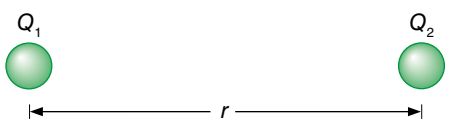


Fig g

(a) Draw F in Figure g.

(b) Find the value of r.

(c) Q_2 is moved to distance $0.1r$ from Q_1. Find the magnitude of the electrostatic force acting on Q_1 by Q_2.

★**8** Point W is 4 cm from a point charge $+4Q$ (Fig h).

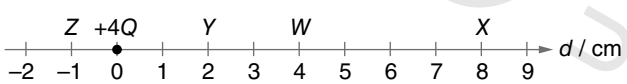


Fig h

If a point charge q is placed at W, it experiences an electrostatic force F. What is the magnitude of the electrostatic force experienced by q if it is placed at

(a) X,

(b) Y,

(c) Z?

★**9** Three point charges X ($+q$), Y ($+q$) and Z ($-5q$) are arranged as shown (Fig i). Y is at a distance $4d$ from X and $3d$ from Z. The magnitude of the electrostatic force acting on X by Y is F.

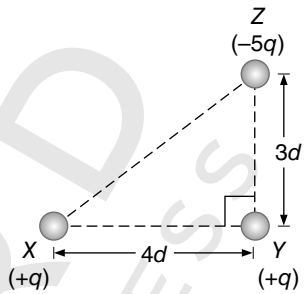


Fig i

Give your answers in terms of F for the following questions.

(a) Find the magnitude of the electrostatic force acting on Y

(i) by X,

(ii) by Z.

(b) Hence find the magnitude of the resultant electrostatic force acting on Y.

★**10** Point charges X, Y and Z are in the region shown in Figure j. They carry the same amount of charge, which is equal to 5×10^{-7} C. Z is positively charged. F_Y and F_Z (**not** drawn to scale) are the electrostatic forces acting on Y and Z respectively by X.

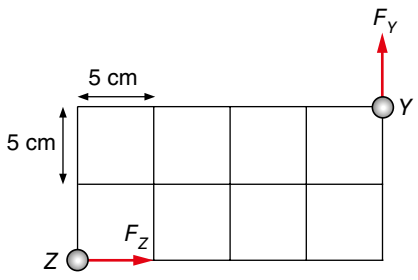


Fig j

(a) Label the position of X in Figure j.

(b) Is Y positively charged or negatively charged?

(c) Find the magnitude of F_Y.

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1.3 Electric field

When a *platypus* dives to hunt, it closes its eyes, ears and nose (Fig 1.3a). How can it locate its prey?

Scientists find that it detects the **electric field** (E-field) produced by its prey from a distance. Let us learn more about the electric field in this part.



Fig 1.3a *A platypus underwater.*

DSE goal

Define electric field strength *E* at a point.

Explain how charges interact with electric field.

1 Electric field strength

Here, instead of 'electrostatic force', 'electric force' is used. The former usually refers to the force exerted between charges while the latter usually refers to the force due to an electric field.

We learned in Book 2 that the gravitational force is a non-contact force that can be studied by using the theory of field. The same applies to electric force. A mass sets up a gravitational field in the space around it. Similarly, a charge sets up an electric field in the space around it (Fig 1.3b).

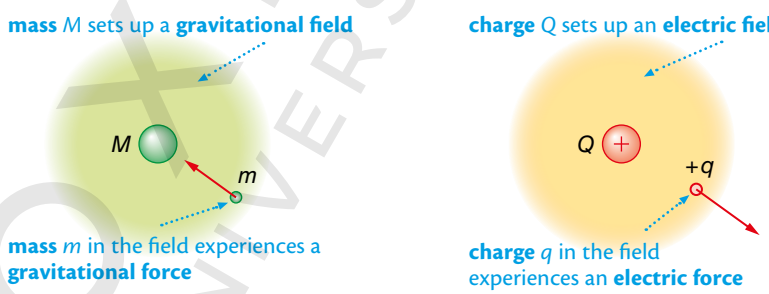
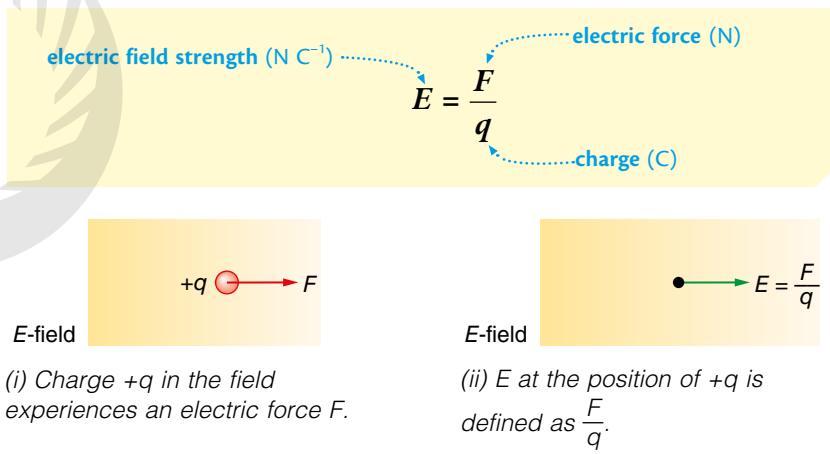


Fig 1.3b *Gravitational field and electric field.*

A test charge only carries a very small amount of charge to avoid disturbing the electric field. It is usually positive.

When a **positive** *test charge q* is placed in an electric field, an electric force *F* acts on it. The **electric field strength** *E* at that point is defined as the electric force per unit charge on *q* (Fig 1.3c).

A test charge carries a positive charge to show the direction of *E* directly.



(i) Charge +q in the field experiences an electric force F.

(ii) E at the position of +q is defined as $\frac{F}{q}$.

Fig 1.3c *Defining the electric field strength.*

platypus 鴨嘴獸　electric field 電場　electric field strength 電場強度　test charge 檢驗電荷

Note that

1. E is a vector. Its direction at a certain point is the same as the electric force acting on a **positive** charge placed at that point.
2. The unit of E is N C^{-1}.
3. E depends on the source producing the electric field but is **independent of the test charge q.**

F depends on q but F/q eliminates the effect of q. Therefore, $E = F/q$ is independent of q.

4. If a negative charge $-q$ is placed in the field, the direction of F is opposite to that of E (Fig 1.3d).



Fig 1.3d *The electric force acting on a negative charge is in the opposite direction of the electric field strength.*

Example 8 Electric field strength at a point

A point charge A of $+2 \times 10^{-8}$ C placed at point X experiences an electric force of 2.4×10^{-3} N pointing to the right.

(a) Find the electric field strength E at X.

(b) If A is replaced with another point charge B of -4×10^{-8} C, find the electric force F acting on B.

Solution

(a) $E = \dfrac{F}{q} = \dfrac{2.4 \times 10^{-3}}{2 \times 10^{-8}} = 1.2 \times 10^{5}$ N C^{-1} (to the right)

(b) $F = Eq = (1.2 \times 10^{5})(4 \times 10^{-8}) = 4.8 \times 10^{-3}$ N (to the left)

Know more

Electric field and gravitational field

The table below compares the electric field and the gravitational field.

	Electric field	Gravitational field
developed by	a charge	a mass
symbol	E	g
definition of field strength	electric force per unit positive charge $E = \dfrac{F}{q}$	gravitational force per unit mass $g = \dfrac{F}{m}$

Quick check 7

→ E-version in OUP Exercise Platform is also available.

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1 A positive test charge q is placed at point X in an electric field. Determine whether the following statements are true.

(a) The electric field strength at X will be larger if the amount of charge in q increases. (T/F)

(b) The electric force acting on q points in the same direction as the electric field strength at X. (T/F)

(c) If q is replaced by a negative test charge, the electric field strength at X will point in the opposite direction. (T/F)

(d) If q is replaced by a negative test charge, the electric force acting on it points in the opposite direction to the electric field strength at X. (T/F)

2 Complete the following table about the magnitude of electric field strength.

	E / N C^{-1}	F / N	q / C
(a)	5000	0.05	
(b)	200		4×10^{-6}
(c)		3×10^{-3}	6×10^{-7}

Ext

DSE goal

State electric field strength around a point charge and solve problems.

DSE exam

13(1B)Q11(b)(i), 15(1A)Q22, 16(1A)Q24, 17(1A)Q23, 21(1A)Q22, 22(1A)Q24

a Around point charges

We are now going to derive a formula for the magnitude of electric field strength E around a point charge. Consider a positive point charge Q and a positive test charge q placed at a distance r from Q (Fig 1.3e).

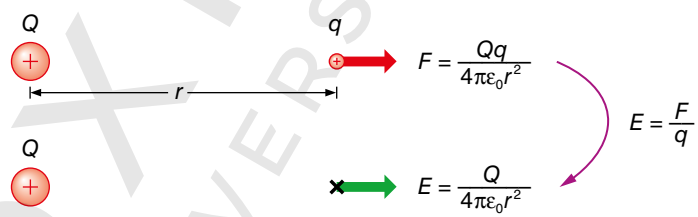


Fig 1.3e *Derivation of the formula for E around a point charge.*

Ts may remind Ss that $E = \frac{F}{q}$ is the definition of electric field strength, while $E = \frac{Q}{4\pi\varepsilon_0 r^2}$ is valid only for a point charge.

electric field strength (N C^{-1}) ····· amount of **charge** (C)

$$E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

permittivity of free space = 8.85 × 10^{-12} C^2 N^{-1} m^{-2} ····· **distance** from charge (m)

From this equation, we can see that E varies inversely with r^2 (Fig 1.3f and 1.3g). ***E* is greater at positions closer to *Q*.**

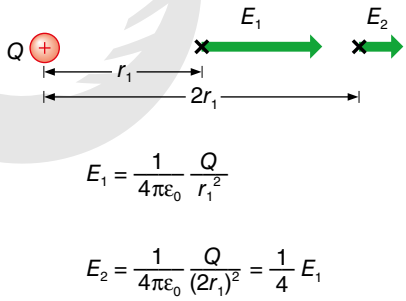


Fig 1.3f *E varies inversely with r^2.*

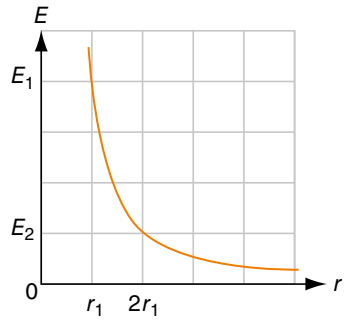


Fig 1.3g *Graph of E against r.*

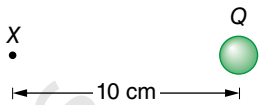
Ext

Example 9 Electric field strength near a point charge

Point X is 10 cm to the left of charge Q. Find the electric field strength at X if Q has a charge of

(a) +10 nC,

(b) –5 nC.



Solution

Tip
When Q is positive, it repels a positive test charge.
When Q is negative, it attracts a positive test charge.

(a) $E = \dfrac{Q}{4\pi\varepsilon_0 r^2} = \dfrac{10 \times 10^{-9}}{4\pi(8.85 \times 10^{-12})(0.1)^2} = 8990 \text{ N C}^{-1}$ (to the left)

(b) $E = \dfrac{Q}{4\pi\varepsilon_0 r^2} = \dfrac{5 \times 10^{-9}}{4\pi(8.85 \times 10^{-12})(0.1)^2} = 4500 \text{ N C}^{-1}$ (to the right)

In general, around a **positive** point charge, the electric field strength **points away** from the charge. Around a **negative** point charge, the electric field strength **points towards** the charge.

Quick check 8

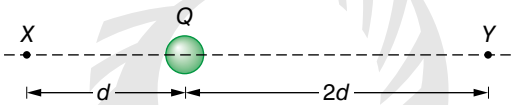
→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Complete the following table about the size of the electric field strength around a point charge. Take $\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$.

	Q	r	E
(a)	2×10^{-6} C	0.5 m	N C^{-1}
(b)	7×10^{-7} C	m	6000 N C^{-1}
(c)	nC	0.03 m	30 000 N C^{-1}

2 X and Y are two points near a point charge Q.



Determine whether the following statements are true.

(a) The magnitude of the electric field strength at X is double that at Y. (T/F)

(b) If the amount of charge in Q is doubled, the magnitude of the electric field strength at X will be doubled. (T/F)

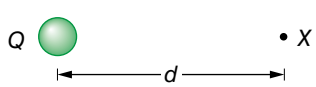
(c) The electric field strengths at X and Y always point in the opposite directions. (T/F)

(d) If Q is a negative charge, the electric field strength at Y will point to the left. (T/F)

Learn by practice 6

Take $\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$.

1 Point X is at a distance d from point charge Q.



(a) If Q has a charge of +8 nC and d is 20 cm, find the electric field strength at X.

(b) If d is 5 cm and the electric field strength at X is 5000 N C^{-1} pointing to the left, find the charge of Q.

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Ext

Simulation

→ The simulation shows the resultant electric field produced by different combinations of charges. It is available on OUP web in PC version (.exe) only.

If a point is under the influence of more than one charge, the resultant electric field strength there is the **vector sum** of the field strengths at that point due to the charges (Fig 1.3h). Note that the resultant electric field strength is zero at point *Y*. This point is called a *neutral point*.

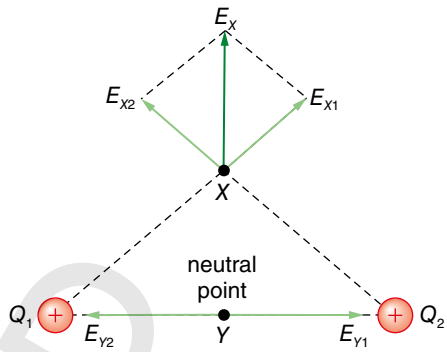
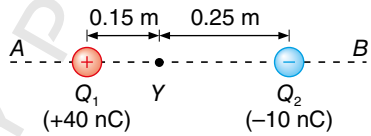


Fig 1.3h *Using vector sum to find the resultant field strengths at points X and Y due to charges Q_1 and Q_2.*

Example 10 Resultant field strength (1-dimension)

Point charges Q_1 (+40 nC) and Q_2 (–10 nC) lie on straight line *AB*. Point *Y* is 0.15 m from Q_1 and 0.25 m from Q_2 as shown.

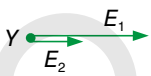


(a) Draw two arrows to show the directions of the electric field strengths E_1 and E_2 at *Y* due to Q_1 and Q_2.

(b) Find the resultant field strength E_R at *Y* due to Q_1 and Q_2.

(c) A neutral point exists on *AB*. What is its exact position?

Solution

(a) 

(b) $E_1 = \dfrac{Q}{4\pi\varepsilon_0 r^2} = \dfrac{40 \times 10^{-9}}{4\pi(8.85 \times 10^{-12})(0.15)^2} = 16\,000$ N C^{-1} (to the right)

$E_2 = \dfrac{Q}{4\pi\varepsilon_0 r^2} = \dfrac{10 \times 10^{-9}}{4\pi(8.85 \times 10^{-12})(0.25)^2} = 1440$ N C^{-1} (to the right)

$E_R = E_1 + E_2 = 16\,000 + 1440 = 17\,400$ N C^{-1} (to the right)

(c) This point must be on the right of Q_2. Let this point be *d* away from Q_2.

magnitude of E_1 = magnitude of E_2

$$\frac{40 \times 10^{-9}}{4\pi\varepsilon_0(0.15 + 0.25 + d)^2} = \frac{10 \times 10^{-9}}{4\pi\varepsilon_0 d^2}$$

$$d = 0.4 \text{ m}$$

The neutral point is 0.4 m to the right of Q_2.

Tip

Between Q_1 and Q_2, E_1 and E_2 always point to the right, so E_R will never be zero.

On the left of Q_1, E_1 is always greater than E_2 in magnitude, so E_R will never be zero.

Think about...

How would the answer of (b) change if Q_2 has a charge of +10 nC instead?

neutral point 中和點

Ext

Skill
Problem-solving

Finding resultant electric field strength

1 Draw the direction of the electric field strength due to each charge.
2 Find the magnitude of each field strength.
3 Resolve the field strengths into components and find the resultant.

→ Teaching notes and exercises are available on OUP web.

Example 11 | Resultant field strength (2-dimension)

Two positive point charges A and B carry the same charge of 1 nC (Fig a). Find the resultant electric field strength E_R at X.

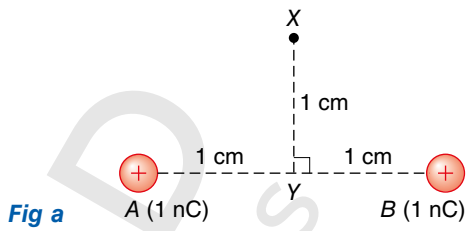


Fig a

Solution

The electric field strength at X due to A and B are as shown (Fig b).

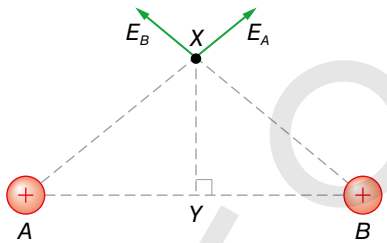


Fig b

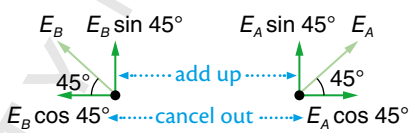


Fig c

Consider the magnitude of E due to each charge.

$r^2 = 0.01^2 + 0.01^2$ ▶

$$E_A = E_B = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{1 \times 10^{-9}}{4\pi(8.85 \times 10^{-12})(0.01^2 + 0.01^2)} = 45\ 000\ \text{N C}^{-1}$$

Resolve E_A and E_B into vertical and horizontal components (Fig c).

The horizontal components cancel each other out. ▶

$$E_R = 2 \times E_A \sin 45° = 2 \times 45\ 000 \sin 45° = 63\ 600\ \text{N C}^{-1} \text{ (upwards)}$$

Quick check 9

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

(For Q1–2.) $PQRS$ is a straight line. Point charges X and Y are placed at points Q and R respectively (Fig a).

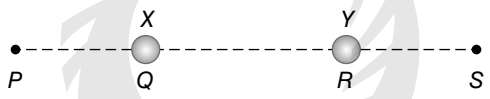


Fig a

1 In which region does the neutral point lie if X and Y are both positive?

A PQ B QR C RS

2 In which region does the neutral point lie if X carries a charge of q and Y carries a charge of $-2q$?

A PQ B QR C RS

3 $ABCDEF$ is a regular hexagon with centre O. A charge $+q$ is placed at A and a charge $-q$ is placed at B (Fig b). What is the direction of the resultant electric field strength at O?

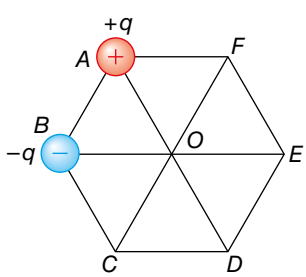


Fig b

A Along OC B Along OD
C Along OE D Along OF

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Ext

Learn by practice 7

1 Two point charges A (+1 nC) and B (–4 nC) are 3 cm apart (Fig a).

Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ N m^2 C^{-2}.



Fig a

Find the resultant electric field strength at points X, Y and Z.

2 Two negative point charges Q_1 and Q_2 have the same charge and are located at the vertices of a square (Fig b). Draw arrows to indicate the direction of the resultant electric field strength at X and Y respectively.

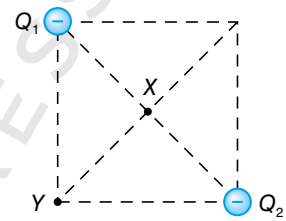


Fig b

b Between parallel charged plates

DSE goal

State electric field strength between parallel plates and solve problems.

DSE exam

14(1A)Q22, 18(1A)Q22, 20(1B)Q9

When two parallel metal plates are connected to the positive and negative terminals of a power supply, they are oppositely charged (Fig 1.3i). It is found that when they are close to each other, the electric field has the **same strength everywhere between the plates** and **points towards the negative plate perpendicularly** (Fig 1.3j). In other words, the electric field between the plates is uniform.

Cross-link

You learned voltage in junior science. We will study it in greater detail in Ch 2.

When the output voltage of the power supply is set to V (i.e. the voltage across the plates is V), and the plates are separated by a distance d, the magnitude of electric field strength E between the plates is:

electric field strength (N C^{-1}) ····· $E = \frac{V}{d}$ ····· voltage across plates (V)

distance between plates (m)

From this equation, E can also be measured in V m^{-1}.

The negative terminal of the power supply is usually earthed. Negative charges are induced on the negative plate.

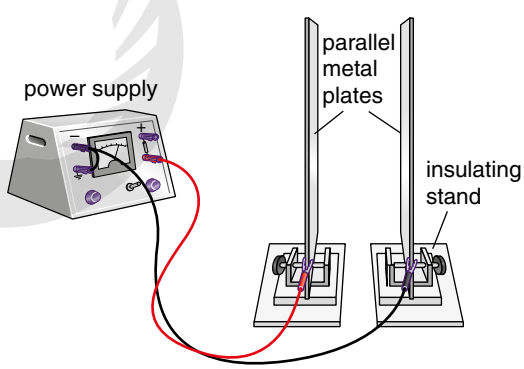


Fig 1.3i Two parallel metal plates connected to a power supply.

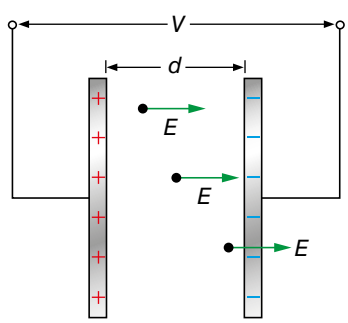
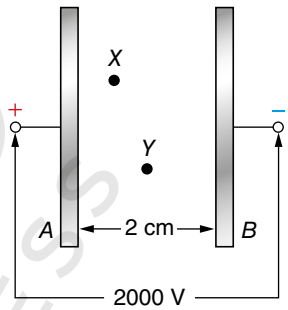


Fig 1.3j E is the same everywhere between two parallel charged plates.

Ext

Example 12 Electric field strength between parallel plates

A voltage of 2000 V is applied across two parallel plates A and B and the distance between them is 2 cm. X and Y are two points between the plates.



(a) Find the electric field strength between the plates.

(b) Find the force acting on a charge q if

(i) it is placed at X and has a charge of +3 nC;

(ii) it is placed at Y and has a charge of –3 nC.

Solution

(a) $E = \frac{V}{d} = \frac{2000}{0.02} = 100\ 000$ N C^{-1} (to the right)

(b) (i) $F = Eq = (100\ 000)\ (3 \times 10^{-9}) = 3 \times 10^{-4}$ N (to the right)

(ii) $F = Eq = (100\ 000)\ (3 \times 10^{-9}) = 3 \times 10^{-4}$ N (to the left)

Tip

E is the same at X and Y.

For a negative charge, E and F point in opposite directions.

Quick check 10

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Complete the following table about the electric field strength between two parallel charged plates.

	E	V	d
(a)	N C^{-1}	5000 V	0.01 m
(b)	20 000 N C^{-1}	1 kV	m
(c)	1.5×10^5 N C^{-1}	V	2 cm

2 Determine whether the following statements about the electric field strength E between two parallel charged plates are correct.

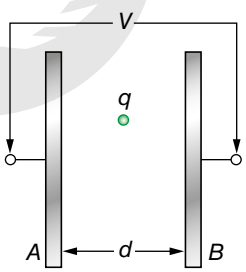
(a) E is greater at positions closer to the positive plate. (T/F)

(b) E points normally to the negative plate. (T/F)

(c) E is directly proportional to the separation between the plates. (T/F)

Learn by practice 8

1 A charge q is placed between two parallel charged plates A and B as shown. The voltage across the plates is V and the distance between them is d.



(a) If the charge of q is +10 nC and the electric force acting on it is 3×10^{-4} N to the left, what is the electric field strength between the plates? Which plate is positive?

(b) If the electric field strength between the plates is 60 000 N C^{-1} towards the left and d is 3 cm, find V and state which plate is positive.

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DSE goal

Describe electric field around a point charge and between parallel plates.

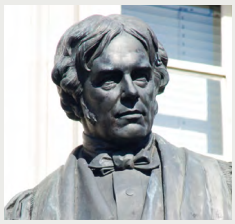
Represent electric field using field lines.

DSE exam

23(1A)Q24

Know more

Michael Faraday (1791–1867)



Michael Faraday was an English chemist and physicist. He made significant contributions to the study of *electromagnetism*.

OUP eResources

2 Electric field patterns

The concept of electric field was developed by *Michael Faraday*. To visualize both the direction and the strength of an electric field, he used **imaginary** lines with arrows called **field lines** to represent the field.

a Around an isolated point charge

Figure 1.3k shows the field lines around an isolated point charge. The field lines point **radially outwards** from the positive charge and **radially inwards** towards the negative charge.

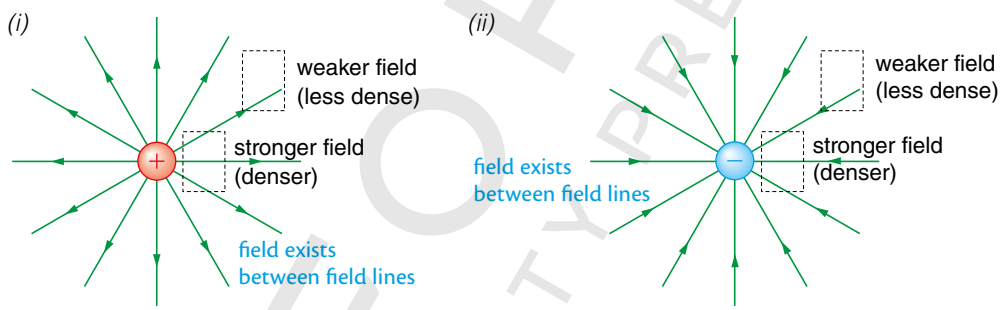


Fig 1.3k *Electric field lines around an isolated (i) positive, (ii) negative charge.*

Note the following points which are common to all electric field patterns:

1. The arrow on a field line indicates the direction of the electric field.
2. Field lines start from positive charges and end at negative charges.
3. The denser the field lines, the stronger the field.
4. Field lines never cross each other.

Recall the earlier discussion that E is greater at a position closer to a point charge.

Otherwise, a test charge at the crossing point would experience a force pointing in two directions, which is impossible.

b Between two parallel charged plates

Figure 1.3l shows the field lines between two oppositely charged parallel metal plates. The field lines point from the positive plate to the negative plate. Since the electric field is uniform between the plates, the field lines are evenly spaced.

Note that the charges on the plates are uniformly distributed.

The field near the edge of the plates and that outside the plates are out of the scope of this book.

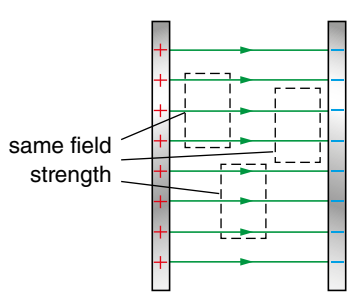


Fig 1.3l *Electric field lines between two oppositely charged parallel plates.*

Michael Faraday 法拉第　　field line 場力線　　electromagnetism 電磁學

C Around two point charges

In the figure below, *X* is a point on the curve. A tangent to the curve at *X* is a straight line that touches the curve at that point.

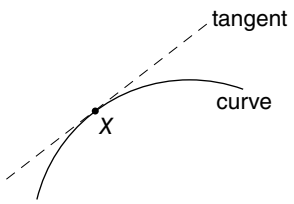


Figure 1.3m shows the field lines around a positive and a negative point charge. The tangent to the field line at a point shows the direction of the field at that point.

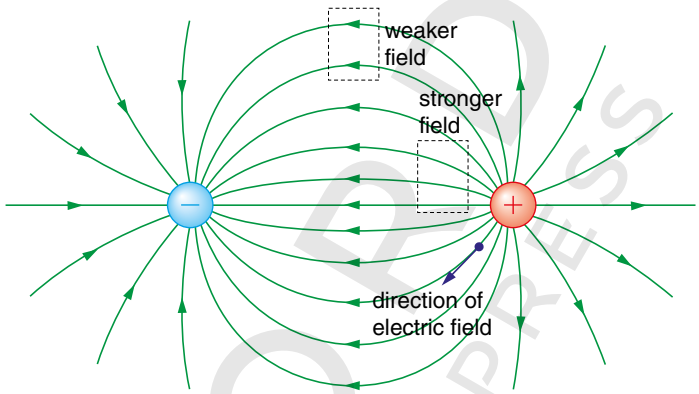


Fig 1.3m *Electric field lines between a pair of unlike point charges.*

Figure 1.3n shows two other cases. A neutral point exists somewhere between two like charges.

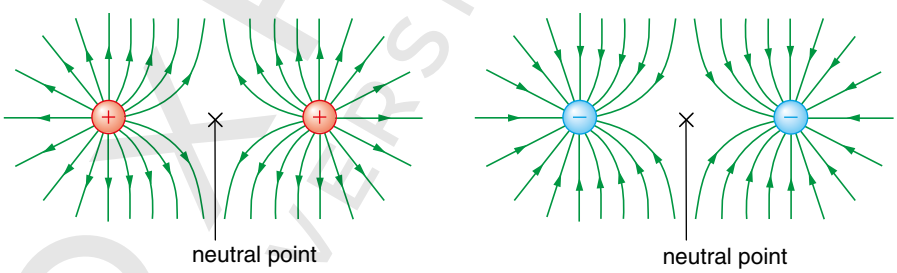


Fig 1.3n *Electric field lines between two equal point charges.*

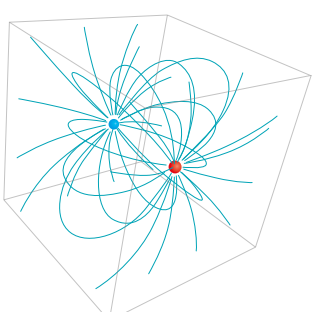


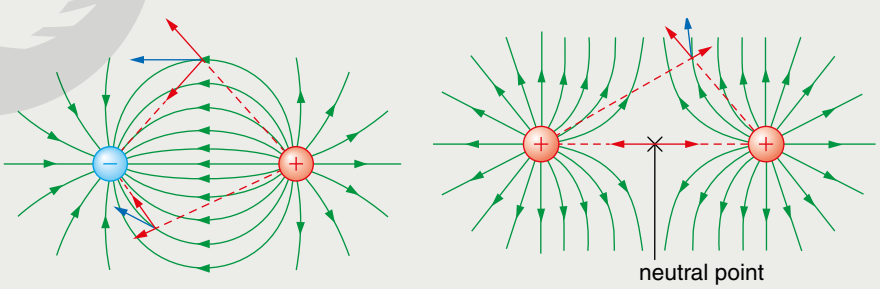
Fig 1.3o *3-D electric field pattern of two unlike point charges.*

Although the field patterns shown in this book are usually two-dimensional, they are actually three-dimensional (Fig 1.3o).

Know more

Electric field patterns of two point charges

At each point around two point charges, the resultant electric field is the vector sum of the fields set up by the two point charges. This results in curved electric field lines in the electric field pattern.



d Seeing the field patterns

Although we cannot see the electric field lines, we can do the following experiment to visualize the electric field patterns. The electrodes are charged when they are connected to the power supply. The *semolina* shows the arrangement of the electric field lines.

→ This video demonstrates Expt 1b.

Video

- Do not handle electrical appliances when your hands, feet or bodies are wet or when standing on a wet floor.
- The floor should be covered with effective insulating materials when conducting this experiment.
- Always connect the negative terminal to the earth terminal of an EHT power supply unless otherwise instructed.
- Always switch off the EHT power supply before making adjustments to the set-up.

Experiment 1b / Electric field patterns

1 Set up the electric field apparatus (Fig a). Pour *castor oil* into a dish and dip a pair of electrodes in the oil.

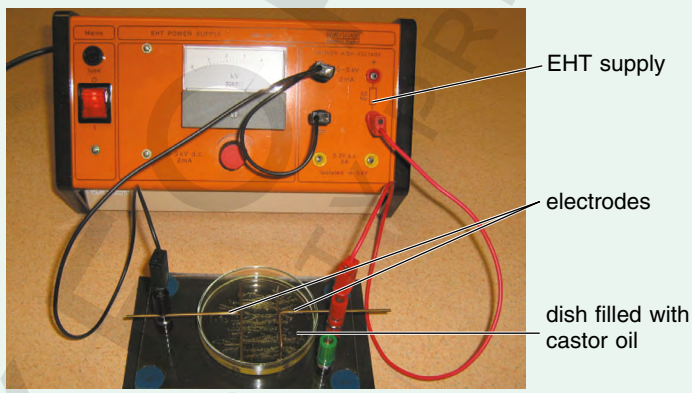


Fig a

2 Connect the electrodes to the + and – terminals of the EHT supply.

3 Sprinkle some semolina onto the oil. Switch on the supply and set the voltage to 5 kV. Observe that the semolina grains line up to show the electric field pattern. Turn off the power supply.

4 Repeat using electrodes of other shapes (Fig b).

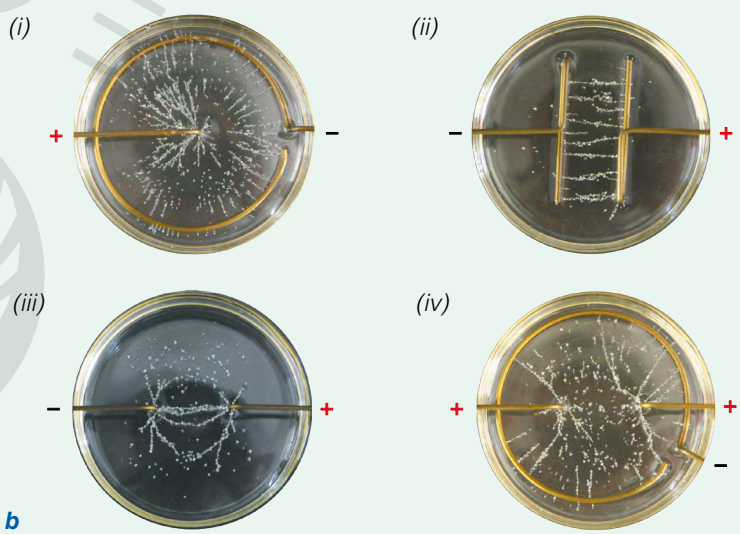


Fig b

Discussion

Does the semolina pattern show the directions of the electric field lines?

Know more

Charging by EHT

An *extra high tension (EHT) power supply* can provide a voltage up to 5000 V. It can charge a conductor quickly. When a conductor is connected to the terminal of an EHT supply, electrons flow into or are drawn from the conductor, making it charged.



semolina 粗粒小麥粉　　castor oil 蓖麻油　　extra high tension (EHT) power supply 超高電壓電源

Know more

Electric potential energy

Do you know how the energy of a charge changes when it moves in a uniform electric field?

▸ p.54

Let us summarize the electric field around a point charge and between two parallel charged plates.

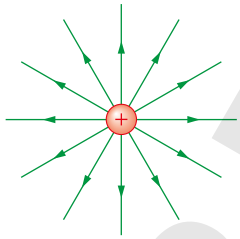
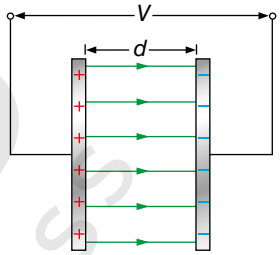
	Point charge	Parallel charged plates
field pattern	 radial straight lines	 uniformly separated straight lines
electric field strength	$E = \frac{Q}{4\pi\varepsilon_0 r^2}$ stronger at position closer to the charge	$E = \frac{V}{d}$ same everywhere between the plates
force acting on charge q	$F = \frac{Qq}{4\pi\varepsilon_0 r^2}$	$F = \frac{Vq}{d}$

Table 1.3a *Electric field around a point charge and between two parallel charged plates.*

Quick check 11

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

(For Q1–2.) The electric field pattern around two point charges P and Q is as shown (Fig a).

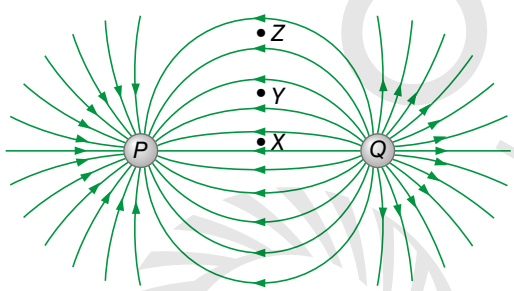


Fig a

1 Determine whether the following statements are true.

(a) P and Q carry opposite charges. (T/F)

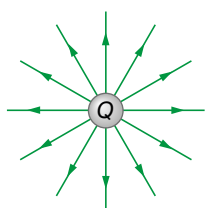
(b) Q must be positively charged. (T/F)

(c) The electric field at points X, Y, and Z are equally strong. (T/F)

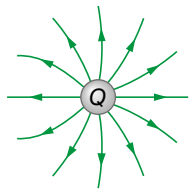
(d) The field lines may cross at a certain point. (T/F)

2 Which of the following diagrams shows the correct electric field pattern if P is removed?

A



B



3 The electric field pattern between two oppositely charged parallel plates P and Q is as shown (Fig b). Determine whether the following statements are true.

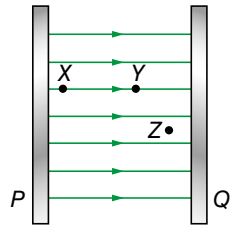


Fig b

(a) P is positively charged. (T/F)

(b) The electric field at X is stronger than that at Y. (T/F)

(c) The electric force acting on a negative charge at Y points towards Q. (T/F)

☆ HKDSE 2013 Paper 1B Q11

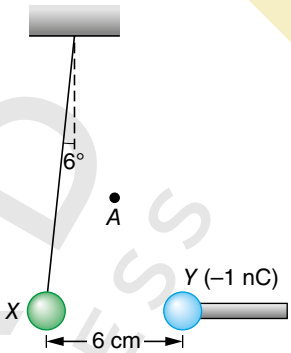
→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Charged sphere suspended by a thread

A small sphere X of mass 0.01 g is suspended by an insulating thread. Sphere Y of charge –1 nC is brought close to X such that the thread suspending X makes an angle of 6° to the vertical. X and Y are 6 cm apart and at the same height. Point A is equidistant from X and Y.

Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ N m^2 C^{-2} and $g = 9.81$ m s^{-2}.



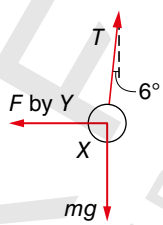
(a) Find the charge of X. (3 marks)

Ext (b) Sketch the direction of the resultant electric field strength due to X and Y at point A. (1 mark)

(c) How will the distance between X and Y change if a neutral metal ball B is placed at point A? (1 mark)

Solution

(a) Consider the forces acting on X.



$T \cos 6° = mg$ ..... (1) 1M

$T \sin 6° = \frac{Q_X Q_Y}{4\pi\varepsilon_0 r^2}$ ..... (2) 1M

(2) ÷ (1):

$$\tan 6° = \frac{1}{4\pi\varepsilon_0} \times \frac{Q_X Q_Y}{mgr^2}$$

$$= (9 \times 10^9) \frac{Q_X(1 \times 10^{-9})}{(0.01 \times 10^{-3})(9.81)(0.06^2)}$$

$$Q_X = 4.12 \times 10^{-9} \text{ C}$$

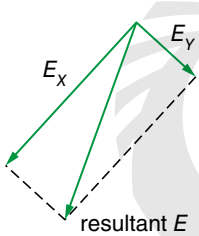
$$= 4.12 \text{ nC}$$

Since X and Y repel each other, X must carry a negative charge.

∴ The charge of X is –4.12 nC. 1A

Common mistake
Some students wrongly halve the electric force. Some wrongly take $4\pi\varepsilon_0 = 9 \times 10^9$ N m^2 C^{-2} or r = 3 cm in substitution. Some draw the free-body diagram incorrectly.

(b)

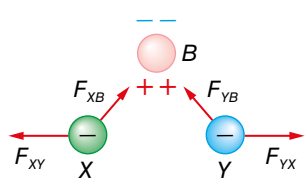


1A

Common mistake
Some students wrongly sketch the whole electric field. Some wrongly draw the field pointing away from the spheres. Some are not aware that $E_X > E_Y$.

(c) Decrease 1A

Common mistake
Students may not know that B attracts X and Y, so the net horizontal force is reduced.



▶ Revision 1 Q29 (p.52)

☆ HKDSE 2018 Paper 1A Q22, 2020 Paper 1B Q9

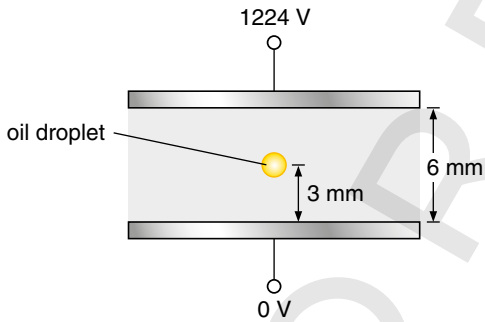
→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Floating charged particle

Robert Millikan measured the charge carried by an oil droplet by suspending the droplet between two horizontal charged plates. Suppose a voltage of 1224 V is applied across the two plates which are 6 mm apart. The mass of the droplet is 2×10^{-14} kg. It is 3 mm above the lower plate.

Take $g = 9.81$ m s^{-2}.



(a) Sketch the field lines between the plates. (2 marks)

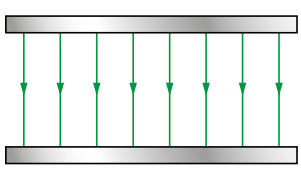
Ext (b) Find the electric field strength between the plates. (2 marks)

(c) Is the oil droplet positively or negatively charged? (1 mark)

(d) What is the charge of the droplet? (2 marks)

Solution

(a)



(Correct direction) 1A

(Perpendicular to plates and evenly spaced) 1A

Common mistake
Students may not be able to draw evenly spaced parallel field lines.

(b) $E = \dfrac{V}{d}$

$= \dfrac{1224}{6 \times 10^{-3}}$ 1M

$= 2.04 \times 10^{5}$ N C^{-1} 1A

Common mistake
Students may mistake the unit of E.

(c) Negatively charged 1A

No net force acts on the droplet. Since the electric field points downwards, the droplet must carry negative charge so that the upward electric force can balance its weight.

(d) By $W = mg$ and $F = Eq$,

$mg = Eq$

$(2 \times 10^{-14})(9.81) = (2.04 \times 10^{5})q$ 1M

$q = 9.62 \times 10^{-19}$ C 1A

Common mistake
Students may wrongly use the equation for point charge.

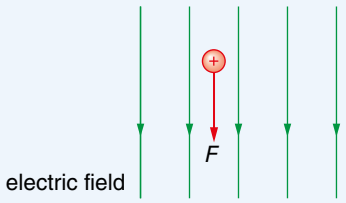
∴ The charge of the droplet is -9.62×10^{-19} C.

▶ Revision 1 Q17 (p.48), Q30 (p.52)

Right and wrong

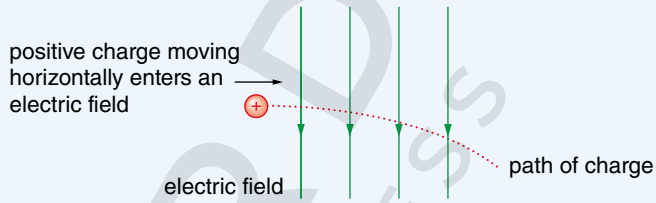
✗ Electric field only exists along the field lines drawn.

✓ Electric field exists in the regions between field lines (except at neutral points). In other words, a charge placed in such a region can experience an electric force.



✗ A charge always moves along an electric field line.

✓ A field line shows the direction of electric force acting on a positive charge but not its moving direction. For example, the charge shown below will move along a parabolic path after entering the field.



Exercise 1.3

If necessary, take $\varepsilon_0 = 8.85 \times 10^{-12}$ C^2 N^{-1} m^{-2} and $g = 9.81$ m s^{-2}.

★1 Ext Three point charges Q_1, Q_2 and Q_3 are situated at the vertices of a square as shown (Fig a). They carry the same amount of charges.

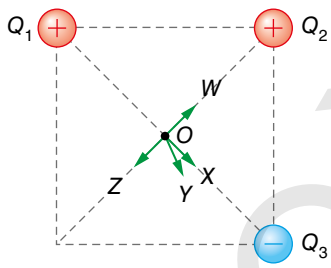


Fig a

Which arrow gives a possible direction of the resultant electric field at the centre O of the square?

A W B X

C Y D Z

★2 An electron moves in an electric field. If forces except the electric force can be ignored, which of the following statements must be correct?

(1) Its moving direction is always parallel to the field.

(2) The electric force acting on it and the electric field point in opposite directions.

(3) It accelerates.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★3 Ext A charge Q is located at the corner of a square (Fig b). If a test charge q is placed at Y, it will experience an electrostatic force F. What is the magnitude of the electric field strength at X?

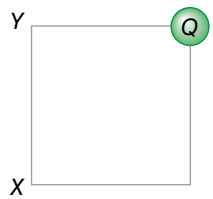


Fig b

A $\frac{F}{2Q}$ B $\frac{F}{2q}$

C $\frac{F}{\sqrt{2}Q}$ D $\frac{F}{\sqrt{2}q}$

★4 Ext Two electrons P and Q are placed between two oppositely charged parallel plates (Fig c). The forces that P and Q acting on each other are negligible.

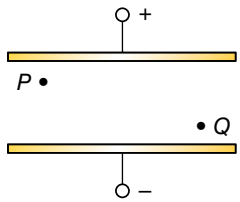


Fig c

Which of the following statements are correct?

(1) The electric force acting on P points upwards.

(2) The electric forces acting on P and Q are the same.

(3) The electric force acting on P will increase if the separation between the plates decreases.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★5 Ext In Figure d, Q_1 carries a charge of +3 nC and Q_2 carries a charge of –5 nC.

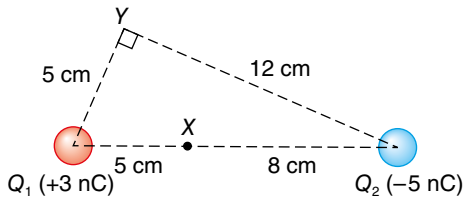


Fig d

(a) Find the magnitude of the resultant electric field strength due to Q_1 and Q_2 at points X and Y.

(b) Where is the neutral point?

★6 Ext Charges Q_1 (+10 nC) and Q_2 are 3 cm apart. The electric force acting on Q_1 by Q_2 is 0.0005 N towards Q_2.

(a) Find the electric field strength at the position of Q_1 due to Q_2.

(b) Find the electric force acting on Q_2.

(c) Find the charge of Q_2.

(d) Find the electric field strength at the position of Q_2 due to Q_1.

★7 Ext A ball of charge +20 nC is suspended by an insulating string between two parallel metal plates that are separated by 5 cm (Fig e). The string makes an angle θ with the vertical. The voltage across the metal plates is 1500 V. The mass of the ball is 0.5 g.

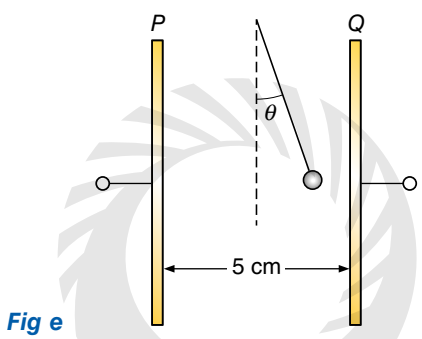


Fig e

(a) Which plate, P or Q, is positive?

(b) Find the electric force acting on the ball.

(c) Find θ.

(d) How would θ change if

(i) the distance between the plates increases;

(ii) the voltage across the plates increases?

★8 Ext Two identical negative charges are placed as shown (Fig f). X is the mid-point between the charges.



Fig f

(a) Sketch the electric field lines around the charges.

(b) State the direction of the electric field strength at X and Y.

★9 Ext Q_1, Q_2 and Q_3 are three identical point charges each carrying the same charge $+q$ (Fig g). They are equally spaced on a straight line L. X and Y are two points on L.

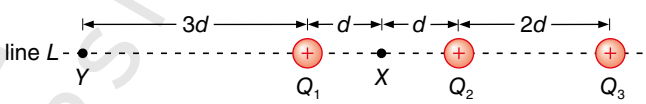


Fig g

(a) Consider the electric field strengths at X and Y.

(i) Are they in the same direction?

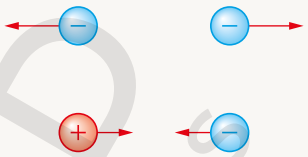
(ii) Which field strength is larger?

(b) Does any position between Q_1 and Q_2 have zero electric field strength? Sketch this position in Figure g and mark it as P if there is any.

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Key concepts 1

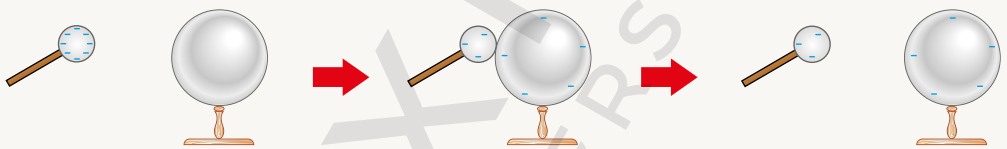
1 Electric charges



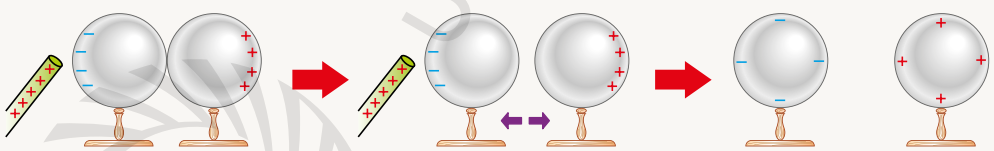
- Two kinds of charges, like charges repel; unlike charges attract
- SI unit of charge: coulomb (C)
- Conductors: allow charges to flow through easily
  Insulator: do not allow charges to flow through

2 Charging and discharging

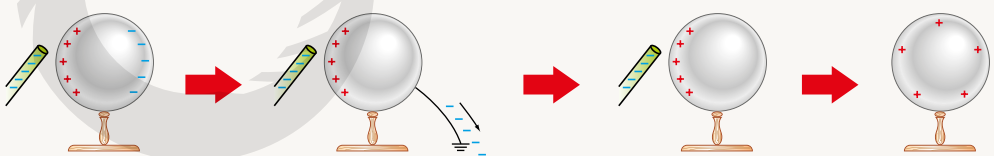
- Charging by friction
- Charging by sharing



- Earthing: connecting a conductor to the Earth; in the absence of other charged objects, a charged conductor becomes discharged when it is earthed.
- Charging by induction



- Charging by induction and earthing



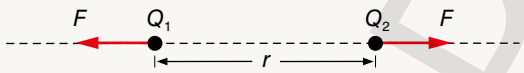
Q29 Exam Report: In (a), although many candidates were able to quote the formula for calculating the electrostatic force, common mistakes included halving the electrostatic force, wrongly taking $4\pi\varepsilon_0 = 9 \times 10^9$ N m^2 C^{-2} or $r = 5$ cm in substitution, drawing incorrect free-body diagrams, etc. In (b)(i), quite a number of candidates omitted this part. Some sketched the whole electric field pattern while a few wrongly identified the resultant field was pointing upwards to the positive charges. In (b)(ii), less than half of the candidates answered this item correctly. This suggests that many candidates failed to understand the induction of charges on a conductor, and the effect of nearby charges or an electric field.

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3

Coulomb's law

Describes the magnitude of electrostatic force between two charges Q_1 and Q_2

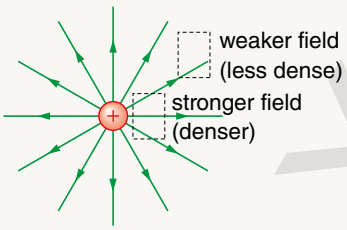


$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$$

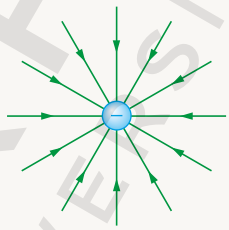
4

Electric field

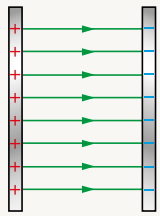
- Set up by charged objects
- Represented by field lines



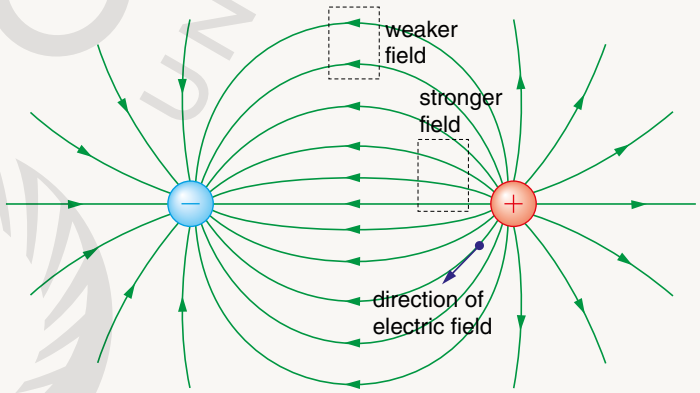
radially outwards from a positive charge



radially inwards towards a negative charge



parallel, evenly spaced and from positive plate to negative plate



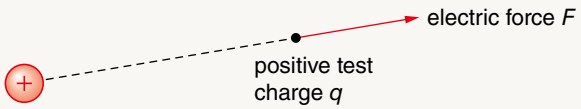
from positive charge to negative charge

- Field lines show the direction of electric force acting on a positive charge
- Field lines start from positive charges and end at negative charges
- A region with denser field lines represents a stronger field
- Field lines never cross

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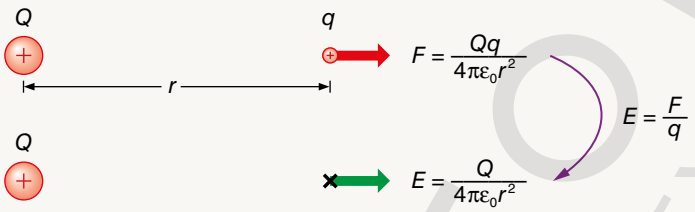
5 Electric field strength *E*

- Force per unit charge on a positive test charge
- Vector, same direction as electric field
- Unit: N C^{-1}



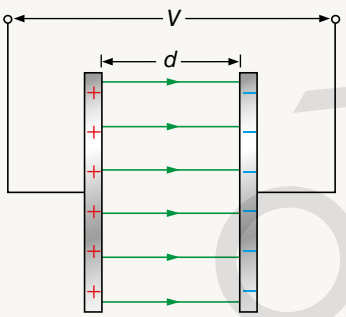
$$E = \frac{F}{q}$$

Ext
- Around a point charge



$$E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

Ext
- Between two parallel charged plates



$$E = \frac{V}{d}$$

Key terms

1	charging 起電	p.8
2	conductor 導體	p.5
3	coulomb (C) 庫倫	p.4
4	Coulomb's law 庫倫定律	p.17
5	discharging 放電	p.8
6	earthing 接地	p.9
7	electric charge 電荷	p.2
8	electric field 電場	p.26
9	electric field strength 電場強度	p.26
10	electric force 電力	p.2
11	electron 電子	p.2
12	electrostatic force 靜電力	p.2
13	electrostatics 靜電學	p.2
14	electrostatic induction 靜電感應	p.5
15	field line 場力線	p.34
16	free electron 自由電子	p.5
17	induced charge 感生電荷	p.5
18	insulator 絕緣體	p.5
19	neutral 中性	p.3

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Revision 1

Multiple-choice questions

1 1.1 Which of the following statements is **incorrect**?

A A charged plastic rod can attract uncharged metal foil.

B Plastic objects can be charged by friction.

C Plastic objects cannot be charged by induction.

D Earthing a charged insulator momentarily can make it become neutral.

***2** 1.1 Two insulated uncharged metal spheres X and Y are placed in contact. A positively charged rod is brought near X (Fig a).

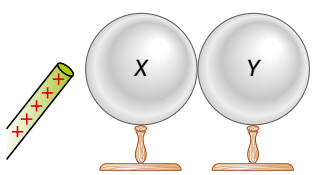


Fig a

What will happen if X is then earthed?

A Electrons will flow to the Earth and sphere X will become neutral.

B Electrons will flow to the spheres and sphere Y will become neutral.

C Electrons and nuclei will flow to the Earth at the same time and both spheres will become neutral.

D Nuclei will flow to the Earth and sphere Y will become neutral.

***3** Ext 1.3 Points X and Y are at distances r and $3r$ respectively from a point charge Q. The electric field strength at X is E. If a test charge q is placed at X, it will experience an electric force F. What is the electric field strength at Y and the electric force experienced by a test charge $2q$ placed at Y?

	Electric field strength	Electric force
A	$\frac{E}{9}$	$\frac{2F}{9}$
B	$\frac{E}{3}$	$\frac{2F}{3}$
C	$3E$	$6F$
D	$9E$	$18F$

***4** 1.3 Figure b shows the electric field established by two point charges.

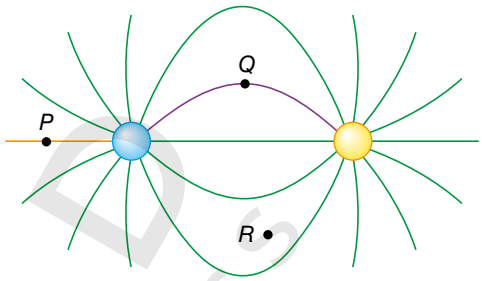


Fig b

Which of the following statements is/are correct?

(1) If a charge is released from rest at P, it will move along the orange field line.

(2) If a charge is released from rest at Q, it will move along the violet field line.

(3) If a charge is released from rest at R, it will not move.

A (1) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

***5** Ext Misc An electron of mass m and charge $-e$ orbits around a nucleus of charge $+e$ in a circular path with a constant speed v (Fig c). The mass of the nucleus is much greater than m. What is the radius of the path of the electron?

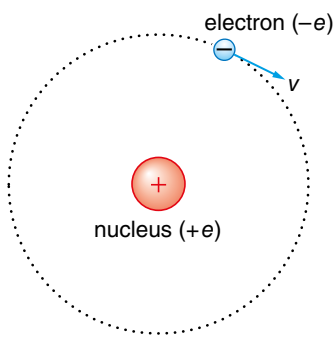


Fig c

A $\frac{4\pi\varepsilon_0 mv^2}{e^2}$

B $\frac{e^2}{4\pi\varepsilon_0 mv^2}$

C $\frac{mv^2e^2}{4\pi\varepsilon_0}$

D $\frac{4\pi\varepsilon_0}{mv^2e^2}$

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*6 Ext Misc A particle of charge $+q$ and mass m is released from rest between two horizontal metal plates (Fig d). The voltage across the plates is V and the distance between the plates is d. The acceleration due to gravity is g. What is the acceleration of the particle?

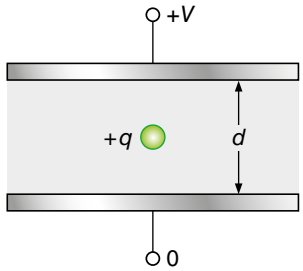


Fig d

A $g + \frac{Vq}{dm}$

B $g + \frac{V}{d}$

C $g - \frac{Vq}{dm}$

D $g - \frac{V}{d}$

*7 Ext Misc Particles X and Y enter an electric field set up by two parallel charged plates in turn at the same speed. The particles have the same mass and size but carry different charges. Their paths are shown in Figure e. Ignore the effect of gravity.

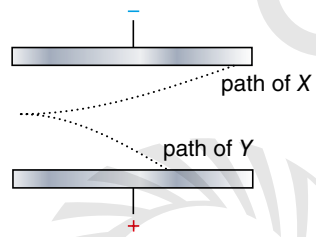


Fig e

Which of the following statements about the particles is/are correct?

(1) X carries a positive charge.

(2) Y experiences a larger electric force than X.

(3) X carries a larger amount of charge than Y.

A (1) only

B (3) only

C (1) and (2) only

D (2) and (3) only

*8 Misc Two small spheres X and Y are suspended by identical insulating string from the same point. They repel each other. At equilibrium, X is at a higher position than Y (Fig f).

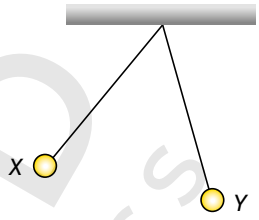


Fig f

Which of the following statements is/are correct?

(1) The force acting on X by Y is larger in magnitude than the force on Y by X.

(2) X may be neutral.

(3) The mass of Y is larger than that of X.

A (1) only

B (3) only

C (1) and (2) only

D (2) and (3) only

*9 1.1 A neutral metal sphere P and a negatively charged metal sphere Q are moved to touch each other. P is then earthed. The spheres are separated afterwards. What are the net charges on P and Q?

	P	***Q***
A	positive	negative
B	negative	negative
C	neutral	negative
D	neutral	neutral

**10 Ext 1.3 In Figure g, point charges Q_1 ($+5 \times 10^{-8}$ C) and Q_2 (-5×10^{-8} C) are 40 cm and 50 cm from point O respectively.

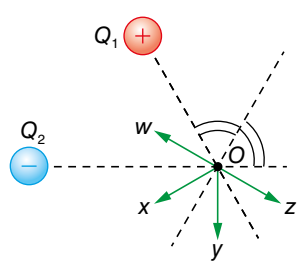


Fig g

Which arrow gives the possible direction of the resultant electric field at O?

A Arrow w

B Arrow x

C Arrow y

D Arrow z

Refer p.31, 38

**11 Ext 1.3 Two positive point charges Q_1 and Q_2 are 4 cm apart, both lying on a straight line L (Fig h). The amount of charge carried by Q_1 is larger than that carried by Q_2.

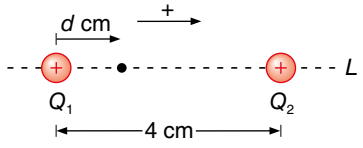
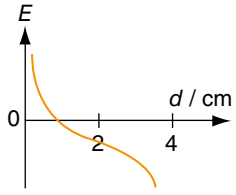
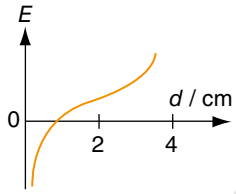
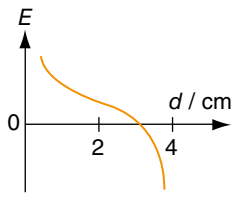


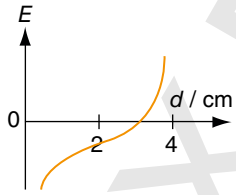
Fig h

Which of the following graphs best describes the resultant electric field strength E at different points on L between Q_1 and Q_2? Take the direction to the right as positive and d as the distance from Q_1.

A 

B 

C 

D 

Refer Eg 10 (p.30)

**12 Ext 1.3



Fig i

Point charges Q_1 and Q_2 are fixed on a straight line passing through points X and Y (Fig i). The resultant electric field due to Q_1 and Q_2 points left at X and points right at Y. Which of the following statements are correct?

(1) The resultant electric field strength is zero somewhere along line XY.

(2) Q_1 is a positive charge and Q_2 a negative charge.

(3) Q_2 is greater in charge quantity than Q_1.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

Refer Eg 10 (p.30)

**13 Misc X and Y are two identical small metal spheres. The amount of charges carried by X and Y are in the ratio 1 : 3 but the type of charges carried by each sphere is unknown. The electrostatic force between them is F. X is brought to touch Y and is placed back to its original position afterwards. Which of the following give(s) the possible magnitude of the electrostatic force between X and Y after they touch?

(1) $\frac{F}{3}$ (2) $\frac{4F}{3}$ (3) $4F$

A (2) only

B (3) only

C (1) and (2) only

D (1) and (3) only

Refer Eg 1 (p.9)

14 HKDSE 2014 Paper 1A Q21 1.2

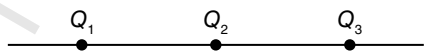


Fig j

Three point charges Q_1, Q_2 and Q_3 are fixed on a straight line with Q_2 at the mid-point of Q_1 and Q_3. The resultant electrostatic force on each charge is zero. Which of the following can be the sign and magnitude (in the same arbitrary units) of Q_1, Q_2 and Q_3?

	Q_1	Q_2	Q_3	
A	+2	+1	+2	(11%)
B	+2	−1	+2	(28%)
C	−4	+1	+4	(18%)
D	−4	+1	−4	(43%)

15 HKDSE 2016 Paper 1A Q24 Ext 1.3

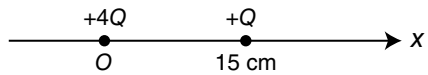


Fig k

Point charges $+4Q$ and $+Q$ are fixed on the x-axis with $+4Q$ at the origin O and $+Q$ at x = 15 cm as shown (Fig k). The respective electric fields due to the two charges are equal at

A x = 10 cm. (34%)

B x = 12 cm. (20%)

C x = 20 cm. (16%)

D x = 30 cm. (30%)

Q14 Exam Report: More than one third of the candidates overlooked the fact that the electrostatic force between two charges is inversely proportional to their separation.

16 HKDSE 2017 Paper 1A Q23

Ext 1.3

A point charge $+Q$ is fixed at a distance d away from the origin O as shown (Fig l).

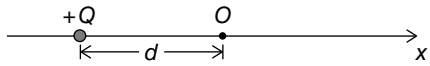
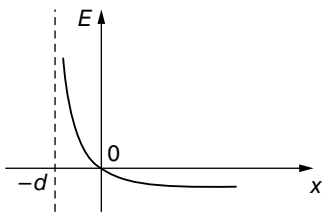


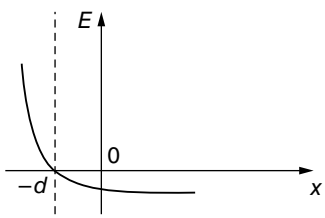
Fig l

Which of the following graphs best represents the variation of the electric field strength E along the x-axis? (Take the electric field pointing to the right as positive.)

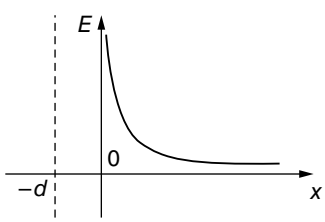
A



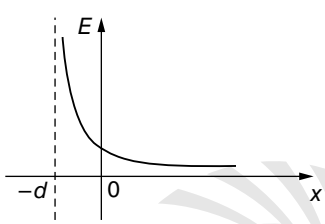
B



C



(55%) D



17 HKDSE 2018 Paper 1A Q22

Ext 1.3

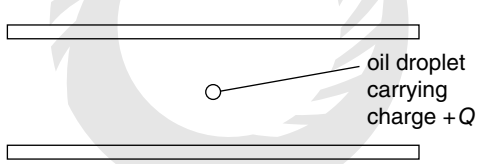


Fig m

A charged oil droplet of mass m is found floating between two horizontal parallel metal plates with an electric field of constant strength in between (Fig m). The oil droplet is carrying charge $+Q$. What are the direction and strength of the electric field?

	Direction	Strength
(55%) A	upward	$\frac{mg}{Q}$
B	upward	$\frac{Q}{mg}$
C	downward	$\frac{mg}{Q}$
D	downward	$\frac{Q}{mg}$

18 HKDSE 2019 Paper 1A Q23

1.2

In Figure n, point charge Y is placed in the middle of two identical positive point charges X and Z, with Z being fixed. Both X and Y are in equilibrium and at rest initially.

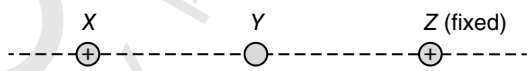


Fig n

What would happen to X if Y is slightly pushed towards Z?

A It moves towards the left. (34%)

B It moves towards the right. (25%)

C It remains at rest. (23%)

D It cannot be determined as the sign of Y is not shown. (18%)

Conventional questions

★**19** Misc

Two small charged spheres are put in a smooth cylinder (Fig o). The charge on sphere X is –3 nC. The spheres have the same mass of 5 g. X floats above Y and remains stationary. The distance between them is 2 cm.

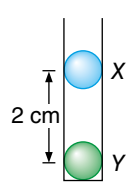


Fig o

(a) Draw a free-body diagram for X. (2 marks)

(b) Find the electrostatic force acting on X. (2 marks)

(c) Find the charge of Y. (2 marks)

(d) Find the electrostatic force acting on Y. (1 mark)

(e) Find the normal force acting on Y by the bottom of the cylinder. (2 marks)

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★ **20** Misc Two small charged spheres A and B suspended by two insulating strings attract each other (Fig p). The spheres are at the same level. The mass of A is 20 g. Assume there are no other charged objects around.

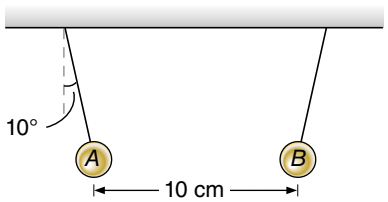


Fig p

(a) Draw a free-body diagram for A. (2 marks)

(b) Find the tension in the string suspending A. (2 marks)

(c) Find the electrostatic force acting on A by B. (2 marks)

(d) The quantity of charge in A is 5×10^{-8} C. Find the quantity of charge in B. (2 marks)

(e) Compare the following quantities.

(i) The electrostatic forces acting on A and B (1 mark)

(ii) Ext The electric field strength at the position of A due to B and at the position of B due to A (1 mark)

★ **21** 1.1 Using the apparatus shown in Figure q and an insulating glove, describe an experiment to illustrate how the size of metal spheres affects the quantities of charge shared by them after they touch each other. You should include the physical quantities to be measured. (6 marks)

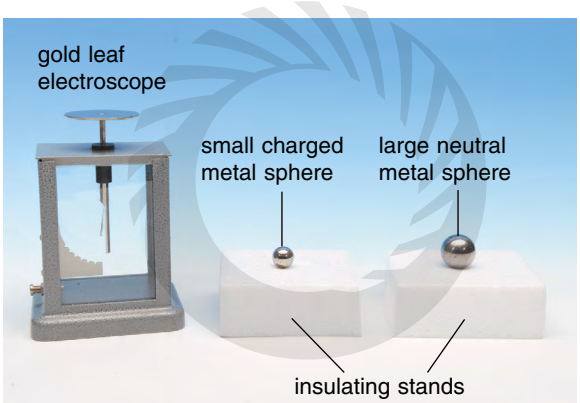


Fig q

★ **22** 1.2 Four point charges P (–1 nC), Q (–2 nC), R (+3 nC) and S (–2 nC) are located at the vertices of a square (Fig r).

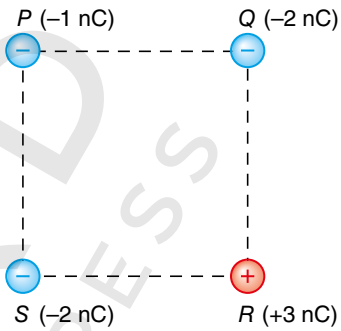


Fig r

(a) Draw arrows in Figure r to show the directions of the electrostatic forces acting on P by Q, R and S. (2 marks)

(b) The resultant electrostatic force acting on P can be reduced to zero by changing the charge of **one** of these point charges.

(i) Which point charge is this? (1 mark)

(ii) Determine its new magnitude of charge. (3 marks)

★ **23** Ext 1.3 Point charges X (–5 nC) and Y (+5 nC) are fixed and are 10 cm apart (Fig s).

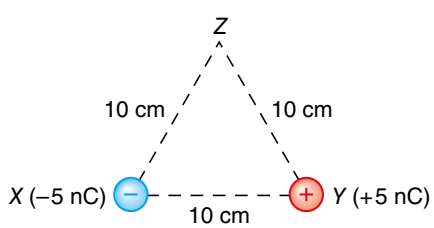


Fig s

(a) Find the resultant electric field strength at point Z. (3 marks)

(b) Another point charge W (–5 nC) is placed at Z. Find the resultant force acting on it by X and Y. (2 marks)

(c) W is moved to somewhere such that the resultant electric field strength at Z becomes zero. Where is W? (2 marks)

(d) W is removed and X is replaced by another point charge Q (+5 nC). If a small neutral sphere is released from rest at Z, what will happen to it? Explain. (4 marks)

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*24 Ext Misc Read the following passage about **pulsed electric field process** and answer the questions that follow.

The pulsed electric field (PEF) process can improve the quality of potato chips. In this process, potatoes are put between two metal plates. High voltage pulses are applied across the plates to produce a very strong electric field (Fig t). Opposite charges are induced on the two sides of each cell in the potatoes. If the electric field is strong enough, the cell wall will be torn slightly and small holes are formed (Fig u). Water molecules can escape through these small holes but starch cannot. These potatoes can produce chips that are crispier and tastier.

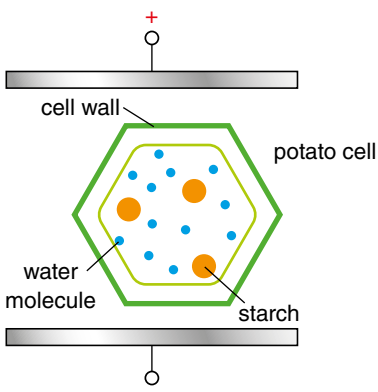


Fig t

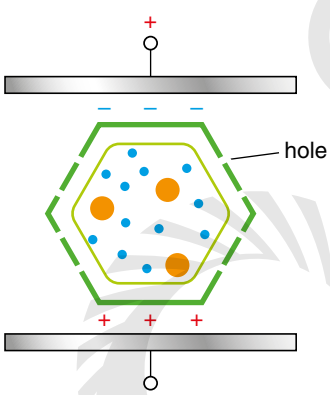


Fig u

(a) Besides increasing the voltage, suggest another way that can increase the electric field strength between the two metal plates. (1 mark)

(b) Why is the cell wall torn when a very strong electric field is applied on the potato? (3 marks)

*25 1.1 Figure v shows a gold leaf electroscope. The metal cap, the metal support and the gold leaf are connected and mounted on an insulated case. The gold leaf can rotate freely about its pivot.

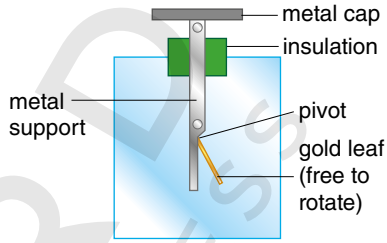


Fig v

(a) The electroscope is not charged and a positively charged rod is put near the metal cap.

(i) Describe and explain the movement of the gold leaf. (3 marks)

(ii) What will happen to the gold leaf when the rod is moved away? (1 mark)

(b) Now, the electroscope is positively charged. What will happen to the gold leaf when

(i) a negatively charged rod is put near the metal cap? (1 mark)

(ii) a neutral iron bar is placed on the metal cap? (1 mark)

**26 Ext 1.3 A neutral metal-coated ball suspended by an insulating thread is placed between two oppositely charged parallel metal plates *P* and *Q* as shown (Fig w). The voltage output of the power supply is 4 kV and plate *Q* is connected to the positive terminal. The separation between the plates is 5 cm. The mass of the ball is 3 g.

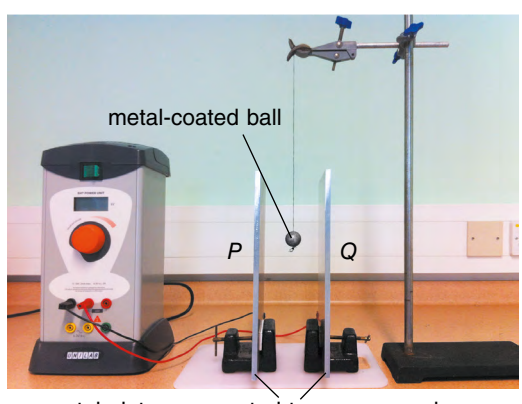


Fig w

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(a) Sketch the electric field pattern between the plates. (2 marks)

(b) Find the electric field strength between the plates. (2 marks)

(c) The ball is touched momentarily by a positively charged metal rod such that it gains +10 nC of charge. The ball does not touch either plate when it deflects.

(i) Find the angle that the insulating thread makes with the vertical. (3 marks)

(ii) The output voltage of the power supply is increased such that the ball touches one of the plates. Describe and explain what happens to the ball afterwards. (4 marks)

Refer p.14, 38

****27** Ext Misc A voltage of 5 kV is applied across two horizontal metal plates which are 4 cm apart. A uniform electric field is set up in region R between them (Fig x). An electron enters the region with a horizontal velocity of 7×10^7 m s^{-1}. The charge and the mass of the electron are -1.60×10^{-19} C and 9.11×10^{-31} kg respectively.

(a) Find the electric field strength in region R. (2 marks)

(b) Find the electric force experienced by the electron in the field. (2 marks)

(c) The electron moves in a path as shown (Fig y).

(i) Find the acceleration of the electron due to the electric field. Explain why the effect of gravity can be ignored when studying the motion of the electron. (3 marks)

(ii) Suppose the electron is 2 cm from the positive plate when it just enters region R. Find the horizontal distance that it travels in region R before hitting the positive plate. (3 marks)

(iii) How will the acceleration of the electron change if the separation of the metal plates increases? (1 mark)

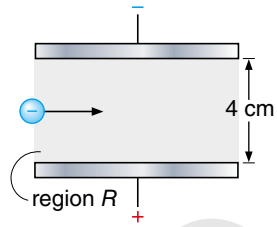


Fig x

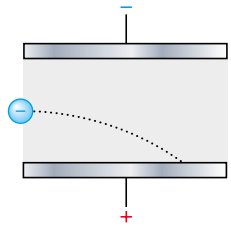


Fig y

Refer p.39

****28** 1.1 In 1752, American scientist Benjamin Franklin invented a device to detect lightning. The device was similar to the one in Figure z.

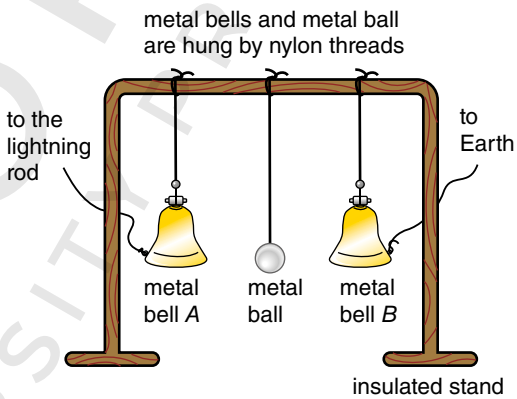


Fig z

When lightning struck the lightning rod on the roof of a house, bell A would be negatively charged. The metal ball would be attracted to bell A, then hit the two bells several times and finally stop.

(a) Explain the motion of the metal ball after it is attracted to bell A. (3 marks)

(b) Suppose the nylon threads are replaced with fixed insulating rods. Lightning strikes the lightning rod. Sketch in Figure aa the charge distribution on bell A, the metal ball and bell B. For simplicity, their shapes are taken as spheres of the same size. (2 marks)

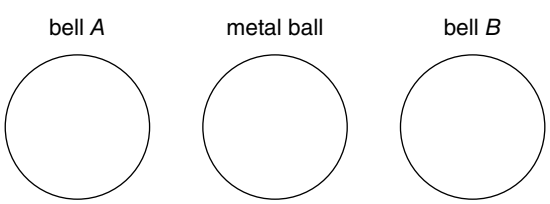


Fig aa

Refer Eg 3 (p.14)

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See p.42 for the Exam Report.

☆ Part (b)(ii) of the original question is omitted.

29 HKDSE 2013 Paper 1B Q11

Misc

Figure ab shows two identical small metal spheres X and Y suspended by insulating threads of the same length. Each sphere has a mass of 1.0×10^{-5} kg and each carries a positive charge of 3.1 nC (1 nC = 10^{-9} C). The separation d of the spheres is 10 cm. The size of the spheres is negligible compared with their separation; therefore they can be treated as point charges.

Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ N m^2 C^{-2}. (g = 9.81 m s^{-2})

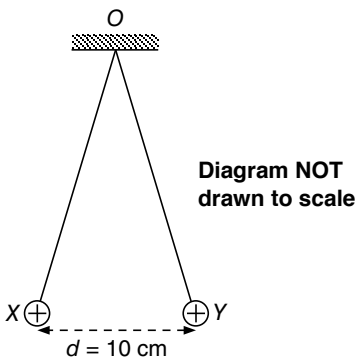


Fig ab

(a) Find the angle between the threads. (3 marks)

Ext (b) Point P is vertically below the fixed point O and it is 10 cm from each sphere (Fig ac).

(i) Indicate the direction of the resultant electric field at P due to these two charged spheres. (1 mark)

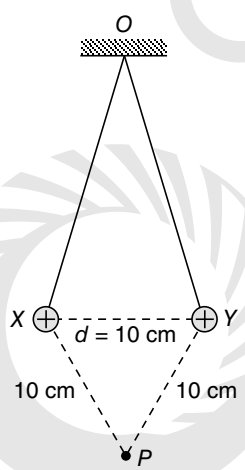


Fig ac

(ii) A neutral metal sphere of finite size is now placed at P. State whether the separation d would increase, decrease or remain unchanged due to the presence of this sphere. (1 mark)

30 HKDSE 2020 Paper 1B Q9

Misc

Two small metal spheres are attached to the ends of an insulating rod of length 5.0 cm. They carry charges $+Q$ and $-Q$ respectively of equal magnitude as shown in Figure ad. The insulating rod is suspended horizontally between two parallel metal plates, A and B, which are connected to an E.H.T. (extra high tension) supply.

The rod is parallel to the metal plates when the E.H.T. is off. After the E.H.T. is switched on, an electric field is set up between the plates and the rod is twisted by an angle of 20° as shown in Figure ae.

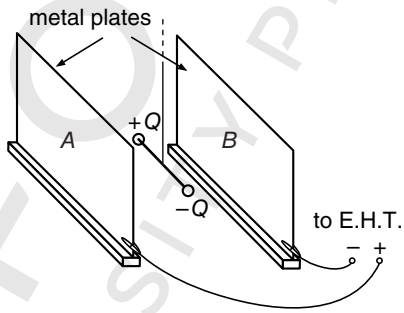


Fig ad

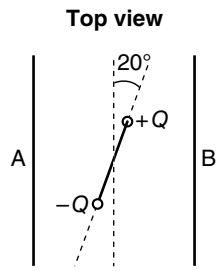


Fig ae

(a) On Figure ae, sketch the electric field lines due to the potential difference across the plates. (2 marks)

(b) The potential difference across A and B is 5.0 kV and the separation between the metal plates is 10 cm. The force due to the electric field acting on each sphere is 2.0×10^{-5} N, find

(i) the moment acting on the rod as shown in Figure ad due to the electric forces on the charged spheres. (2 marks)

Ext (ii) the strength of the electric field E due to the potential difference across the metal plates. (2 marks)

(iii) the magnitude of the charge Q on the spheres. (2 marks)

Q30 Exam Report: Candidates' performance was poor. Although most candidates were able to show the correct direction of the electric field in (a), many failed to draw evenly spaced, parallel field lines properly. In (b)(i), not many were able to use the correct component of force or moment arm to find the moment acting on the rod. Some mistook the length of the rod or even the separation of the plates as the moment arm. Wrong units for the moment such as J or N were common. Most candidates answered (b)(ii) correctly. Incorrect units for the electric field were found occasionally. Many candidates wrongly employed equations of force or electric field due to a point charge to tackle (b)(iii).

Know more

Conductors and insulators (p.5)

1 The charges moving in a conductor may not be electrons. For example, the moving charges in tap water are positive and negative ions.

2 Metals are good conductors. Wood and plastic are good insulators.

3 For most materials, it is not appropriate to simply classify them as conductors or insulators because their ability to allow charges to flow lie somewhere between a good conductor and a good insulator. For example, tap water are fairly good conductors.



4 Materials containing more free electrons are better conductors.

5 You learned about heat conductors in Book 1. A heat conductor is usually a good electrical conductor. However, there are some exceptions. For example, diamond is a heat conductor but an electrical insulator.

STSE

Air quality (p.8)

The *World Health Organization* estimated that about seven million people in the world die of air pollution-related diseases every year. If we inhale a large number of particulates with a diameter of 10 μm or less, our health, especially the *pulmonary function*, may be damaged. To improve air quality, electrostatics are applied in many aspects to reduce *suspended particulates* such as dust and fumes.

Air purifiers are commonly found in homes and offices (Fig a). One of their major functions is to reduce the suspended particulates indoors. A typical type of air purifier applies electrostatics to pick up particulates (Fig b).

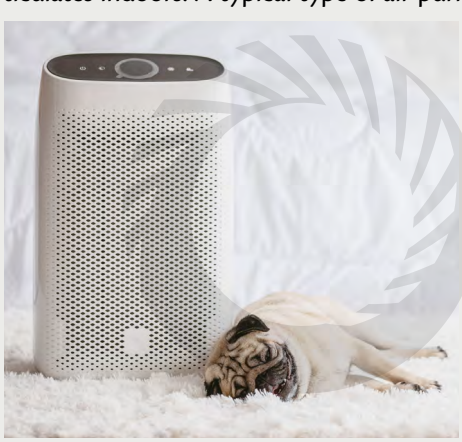


Fig a

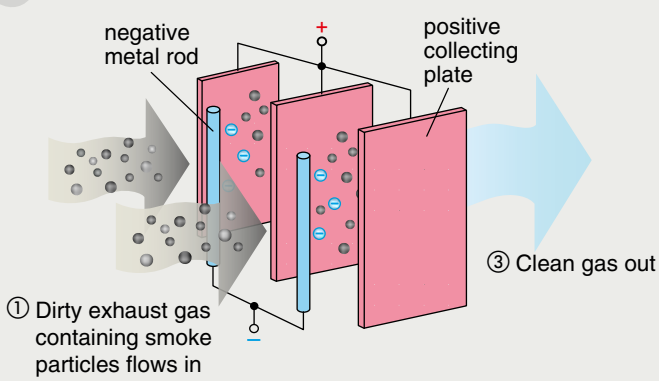


Fig b

Coal-fired power stations also use similar methods to remove the smoke particles from the exhaust gas and prevent the particles from dispersing into our environment.

World Health Organization 世界衛生組織　　pulmonary function 肺功能
suspended particulate 懸浮粒子　　air purifier 空氣清新機

Know more

Electric shock in winter (p.12)

Usually, our clothing and shoe soles are insulators. When we move, static charges may develop on them by friction. The static charges will induce charges in our body. If the amount of charges is large enough, they may jump across the small air gap between our body and another object (Fig a). This results in a spark and thus an electric shock. Lightning (Fig b) is a similar phenomenon but on a much larger scale.



Fig a



Fig b

You can hear or even see sparks when you take off a woollen sweater.

Water vapour in air can take away the static charges on an object. In winter, the humidity is usually low, so static charges are more likely to accumulate and electric shocks happen more easily.

Electric potential energy (p.37)

When an object of mass m is lifted upwards by an external force F through a height h at a constant velocity, work is done on the object against the gravitational force and the object gains gravitational potential energy (Fig c).

Gain in gravitational PE = work done by F = Fs = $(mg)h$ = mgh

When the object is released, the object accelerates under the effect of the gravitational force. Its gravitational PE is converted into its kinetic energy.

Similarly, when a positive charge q is moved by an external force F towards the positive plate through a distance d at a constant velocity, work is done on q against the electrostatic force and q gains *electric potential energy* (Fig d).

Gain in electric PE = work done by F = Fs = $(qE)d$ = qEd

When q is released, it accelerates under the effect of the electrostatic force. Its electric PE is converted into kinetic energy.

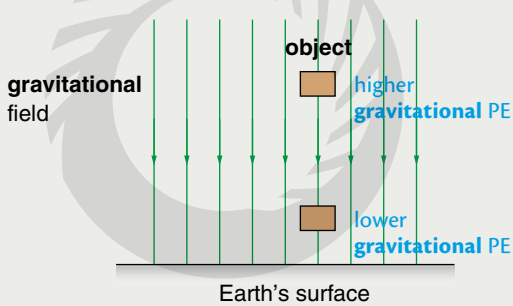


Fig c

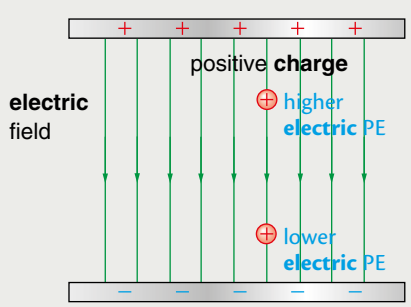


Fig d

electric potential energy 電勢能